Is There an Optimal Decision Theory?

Within-Problem Conflicts, Cross-Problem Axioms, and the Semantics of Menus

MARIO ASIS

October 2026

ABSTRACT. It is often suggested that no decision theory can be optimal on every fair decision problem, because the standards by which theories are judged conflict. This paper makes the claim precise and determines when it holds. On a class of finite problems with prediction, in which a structural representation fixes every causal counterfactual and fairness and well-definedness are separate conditions, a decision theory is any assignment of policies to problems. Ex-ante optimality is then always attainable, so the claim is false as stated. Within a single problem, every inconsistency among ex-ante optimality, structural causal dominance, ratifiability, commitment stability, and an evidential criterion has a core of one or two criteria; the conflict between ex-ante optimality and dominance is possible exactly when the state law depends on the prescribed action, and it survives imperfect prediction and randomisation. Across problems, each problem is reduced to its causal matrix, whose rows are the belief states induced by the agent's policies and whose diagonal is ex-ante value. With common affine invariance the cross-problem axioms again reduce to singleton and pairwise cores, one of which is Briggs's theorem. With invariance under row translations, neutrality and contraction force a rule to choose every option, on exactly the domains satisfying a reachability condition; and principal binary independence, with symmetry, ordering, and monotonicity, characterises ex-ante evaluation. Both results depend on how a smaller menu is modelled: if the belief model is held fixed, baseline-invariant evaluation is menu coherent and contraction no longer forces ex-ante evaluation. The impossibilities concern the preservation of comparisons when the belief perspectives of removed options disappear, not the preservation of choices when options disappear.

CONTENTS

1. Introduction 2

2. Decision problems with prediction 7

3. Optimality criteria and the possibility of ex-ante optimality 9

4. The witness problems 11

5. Commitment, evidence, and the within-problem map 14

6. Robustness of the within-problem conflict 19

7. The causal matrix and cross-problem axioms 22

8. A pairwise reduction under common affine invariance 26

9. Row-baseline invariance and contraction 28

10. The semantics of a smaller menu 33

11. What characterises ex-ante evaluation 38

12. Types, parameters, and certification 41

13. Mixed policies and evidential semantics 43

14. Conclusion 45

A. Register of stipulations 48

References 49

1. Introduction

Is there a decision theory that gets every decision problem right? For half a century the question has been pursued case by case. A theory is proposed; a problem is constructed on which its verdict seems wrong; the theory is amended or a rival proposed; and the rival meets its own counterexample. The impression this history leaves is that no theory can be optimal everywhere, because the standards by which theories are judged pull in different directions. This paper tries to turn that impression into theorems. It asks, for precisely defined criteria of optimality and precisely defined classes of problems, when a decision theory satisfying them exists and when it does not, and it tries to close the question in the sense of determining every inconsistency among the criteria considered and exhibiting a witness for every consistent combination.

1.1. From dominance to Newcomb's problem

The expected-utility theories of Ramsey (1931), von Neumann and Morgenstern (1944), and Savage (1954) model an agent's options as functions from states of the world to consequences, with the states not depending on what the agent does. Under that assumption, the principle of dominance—prefer an option that is better in every state—is a consequence of maximising expected utility, and the two principles never conflict. Jeffrey (1965) replaced the act-independent states with a single algebra of propositions, valuing an act by the news it would bring; Bolker (1966, 1967) supplied the representation theorem. In Jeffrey's theory the probability of a state may depend on the act, and dominance and expected utility can come apart.

Nozick (1969) made the possibility vivid. A reliable predictor has placed a large sum in an opaque box if and only if it predicted that the agent would take only that box; a transparent box contains a small sum. Taking both boxes dominates, since the contents are fixed; taking only the opaque box maximises expected utility conditional on the act. Nozick credited the problem to the physicist William Newcomb and proposed restricting dominance to cases where the states are probabilistically independent of the acts. Gibbard and Harper (1978) gave the conflict its canonical form by distinguishing two expected utilities: one computed with the probabilities of counterfactual conditionals, which measures what an act would causally bring about, and Jeffrey's, which measures what it indicates. Causal decision theory, as the first came to be called, takes both boxes; evidential decision theory takes one. Gibbard and Harper also introduced Death in Damascus, in which Death, having predicted the agent's destination, waits there: whichever city the agent chooses, the other looks better. Lewis (1979) observed that the prisoner's dilemma played against a sufficiently similar partner is a Newcomb problem, and Lewis (1981b) unified the causal theories of Gibbard and Harper, Skyrms (1980, 1982), and Sobel (see Sobel 1994) by means of dependency hypotheses. Against the charge that causal decision theorists end up poorer, Lewis replied that Newcomb's problem rewards irrationality (Lewis 1981a). On the evidential side, the tickle defence of Eells (1982) argued that a self-aware agent's own beliefs and desires screen off the correlations on which one-boxing relies, and Spohn (1977) argued that deliberation crowds out prediction, so that an agent cannot coherently assign probabilities to its own current options. Joyce (1999) gave causal decision theory a representation theorem, and by the end of the century it had become the orthodox view (Weirich 2024).

1.2. Instability, ratifiability, and the counterexamples

Death in Damascus exposed a second fault line. Jeffrey (1981, 1983) proposed that a rational choice should be *ratifiable*: optimal relative to the beliefs the agent would have upon making it. In Death in Damascus no pure option is ratifiable. Harper (1986) proposed maximising causal expected utility among ratifiable options and admitting mixed acts so that such options exist; Richter (1984), Weirich (1985), and Rabinowicz (1988) studied the instability that makes every option look worse once chosen; and Skyrms (1990b) modelled deliberation as a dynamical process whose rest points are deliberational equilibria, connecting ratifiability with equilibrium in games (Skyrms 1990a). Sobel (1994) collected much of this work.

Egan (2007) reopened the debate with cases in which an act is evidence about the very causal structure that determines its outcome—shooting as evidence of a lesion that would make one miss, pressing a button that kills all psychopaths as evidence that one is a psychopath—and in which causal decision theory seems to give the wrong verdict. The responses define the current landscape. Deliberational theories recommend equilibria of a process of self-updating (Arntzenius 2008; Joyce 2012, 2018); benchmark theory assesses each option against a state-dependent benchmark (Wedgwood 2013); Spohn (2012) argued that a properly formulated causal theory one-boxes; Ahmed (2012, 2014a,b) revived the evidential theory with cases in which

causal reasoning fails; Hare and Hedden (2016) classified decisions as self-reinforcing or self-frustrating and criticised the stability-based remedies; Gallow (2020) proposed a causal theory that manages the news an act carries; and Gallow (2026) argues for comparativism, on which instrumental value is fundamentally a matter of pairwise comparison. The argument from wealth was reconsidered as well: Bales (2018) and Spencer and Wells (2019) argued that it compares agents who face different problems, and Wells (2019) constructed a problem with equal opportunities in which causal reasoners predictably end up richer. The collection edited by Ahmed (2018) surveys the state of play.

1.3. Policies, plans, and fairness

A third line of work evaluates not acts but policies. Kuhn's analysis of extensive games introduced perfect recall and the equivalence of mixed and behavioural strategies under it (Kuhn 1953). Hammond (1988) derived expected utility from consequentialism and dynamic consistency in decision trees; McClennen (1990) defended resolute choice, the execution of an ex-ante plan even when it is not optimal ex post; and Gauthier (1986, 1994) argued that dispositions which are rational to adopt can make it rational to carry out assurances and threats. Piccione and Rubinstein (1997) separated planning-stage from action-stage optimality in the absent-minded driver, and Aumann et al. (1997) showed that the planning-optimal strategy is action-optimal although the converse fails; Arntzenius et al. (2004) described sequences of individually rational choices whose sum is disastrous and from which only binding escapes.

In artificial intelligence the policy view became a family of decision theories. Updateless decision theory (Dai 2009) selects the input–output map that is optimal under the agent's prior; timeless decision theory (Yudkowsky 2010) treats a decision as the output of an abstract computation; and Soares and Fallenstein (2015) and Yudkowsky and Soares (2017) developed these ideas into functional decision theory, with a peer-reviewed presentation by Levinstein and Soares (2020). Functional decision theory is claimed to do well on *fair* problems, those in which the outcome depends only on the agent's behaviour in the problem at hand; its authors leave a formal criterion of fairness as future work and suggest that a general-purpose theory may match or outperform all rivals on fair problems if one can be found. MacAskill (2019) objected with the Bomb, a problem in which the policy-optimal action is to take a box known to contain a bomb, violating the principle that under certainty one should choose the best outcome. Oesterheld and Conitzer (2021) showed how to extract money from causal decision theorists by predictive offers; Everitt et al. (2015) formulated sequential, act-based and policy-based versions of the causal and evidential theories; Bell et al. (2021) showed that value-based reinforcement learners in Newcomb-like environments converge, if at all, to ratifiable policies, and that continuous environments always admit one; Halpern (2020) gave conditions for the existence of ratifiable mixed acts; MacDermott et al. (2023) characterised the theories with mechanised causal graphs; and Oesterheld and Conitzer (2024) and Cooper et al. (2024) studied when self-locating beliefs make act-level causal or evidential reasoning agree with the ex-ante optimal policy. Wolpert and Benford (2013) argued that the

two Newcomb arguments presuppose incompatible formalisations of how choice and prediction are related.

1.4. Impossibility results

Three results are of the form this paper pursues. Skalse (2021) proposed a general counterexample to any decision theory: an opaque box that is empty exactly when the target theory permits taking it. Briggs (2010) proved that no completely general decision theory satisfies both a Pareto condition across the evaluations of an agent's possible post-decision selves and a self-sovereignty condition on which the comparison of two options depends only on how those two options evaluate all options, modelling decision-theoretic paradoxes on voting paradoxes and Sen's liberal paradox. Podgorski (2022) proved that, for theories assigning each option an overall value, independence of irrelevant alternatives forces the comparison of two options to depend only on their values by their own lights, and so singles out the evidential theory; his tournament decision theory escapes the result by producing no overall value.

These results draw on the theory of social choice, whose methods the present paper also uses. Arrow (1951) established that no aggregation rule satisfies a handful of individually innocuous conditions on an unrestricted domain. The contraction condition on choice functions goes back to Chernoff (1954); Arrow (1959) showed that choice from all finite subsets is rationalised by an ordering exactly when it satisfies the weak axiom of revealed preference; and Sen (1970, 1971, 1977, 1993) named and analysed the conditions α, β, and γ and the limits of internal consistency of choice. Contraction consistency conflicts with majority choice through the binary cycle, a fact central to the theory of tournament solutions (Brandt et al. 2016; Brandt and Harrenstein 2011; Laslier 1997; Moulin 1986), and it continues to generate impossibilities (Basu Mallik and Bhattacharya 2026; Blair et al. 1976).

1.5. The question and the answer

The thesis this paper set out to prove was that no decision theory is optimal on all fair decision problems, the obstruction coming from mutually incompatible criteria of optimality rather than from any demand for a general rule. To make it precise, Section 2 defines a class of finite problems with prediction in which every counterfactual the criteria need is fixed by a structural representation, and in which fairness and well-definedness are separate conditions. A decision theory is any assignment of policies to problems, lookup tables included, so an impossibility must hold for every such assignment.

The thesis is false as stated. Since every finite problem has finitely many policies, choosing an ex-ante maximiser in each problem is a decision theory that is ex-ante optimal everywhere (Theorem 3.6). What is true is subtler, and the paper establishes it at two levels.

Within a single problem, the conflicts among the natural criteria—ex-ante optimality, structural causal dominance, ratifiability, commitment stability, and an evidential

criterion—are shallow. Every inconsistency has a core of one or two criteria (Theorems 5.7 and 5.10), the central pair being Nozick's. A conflict between ex-ante optimality and dominance is possible exactly when the state law depends on the prescribed action (Theorem 4.2); it survives imperfect prediction and randomisation, with an exact trade-off between the two kinds of regret (Proposition 6.1).

Across problems, each problem is reduced to its causal matrix, whose rows are the belief states induced by the agent's possible policies and whose diagonal is ex-ante value; every real square matrix arises (Theorem 7.2). With common affine invariance, the cross-problem axioms again reduce to singleton and pairwise cores, one of which is Briggs's theorem (Theorem 8.1). With invariance under row translations—the matrix form of evaluating options against the baseline of each belief state—a genuine four-axiom impossibility appears: neutrality, row-translation invariance, and contraction force a rule to choose every option (Theorem 9.3), on exactly those domains satisfying a reachability condition (Theorem 9.7). Ex-ante evaluation, in turn, is characterised by principal binary independence together with neutrality, affine invariance, ordering, non-triviality, and monotonicity (Theorem 11.1 and Corollary 11.2), whose core is Podgorski's theorem.

Both cross-problem results depend on how a smaller menu is modelled. If removing options also removes the beliefs that those options would induce, the results hold. If the belief model is held fixed, row-baseline-invariant evaluation becomes menu coherent, and contraction no longer forces ex-ante evaluation (Section 10 and Theorem 11.4). The dichotomy that the earlier results suggest—menu coherence forces ex-ante evaluation, and causal evaluation cannot be menu coherent without triviality—is therefore false as stated. The precise replacement is the conclusion of the paper: *the impossibilities concern the preservation of comparisons when the belief perspectives associated with removed options disappear, not the preservation of choices when options disappear.*

Several results are known and are presented as such: the Newcomb conflict (Nozick), the absence of ratifiable pure acts (Gibbard and Harper; Jeffrey), mixed existence under continuity (Halpern; Bell and co-authors), the Pareto–self-sovereignty conflict (Briggs), and the core of the binary characterisation (Podgorski). The consistency maps, the characterisation of when dominance and ex-ante optimality can conflict, the trade-off identity, the analysis of noisy commitment, the circulant theorem and its exact domain criterion, the comparison of menu semantics, and the cardinality results on families of rules were not found in the literature searched; their ingredients are classical, and no claim of priority is made.

Sections 2–6 treat single problems: the model, the criteria and the possibility theorem, the witness problems, commitment and the within-problem maps, and robustness. Sections 7–11 treat cross-problem axioms: the causal matrix, the pairwise reduction, row translations and contraction, menu semantics, and the characterisation of ex-ante evaluation. Section 12 asks whether the relevant problem types are few, Section 13 extends the results to mixed policies and two evidential semantics, and Section 14 states the final thesis and the questions left open. Appendix A lists every stipulation made.

2. Decision problems with prediction

This section fixes the class of problems on which decision theories are compared. The aim is a model rich enough to contain the problems that drive the debate (Newcomb's problem, the twin prisoner's dilemma, Death in Damascus, the Bomb, counterfactual mugging) and precise enough that every criterion to be considered has a determinate verdict on every problem. Each place where an informal notion—a counterfactual, "the same computation", "holding the state fixed"—has to be settled by stipulation is marked as such; the stipulations are collected in Appendix A.

2.1. Problems, policies, and structural plans

Definition 2.1 (Decision problem). A *decision problem* D is a finite extensive-form tree with three kinds of non-terminal nodes and real payoffs at its leaves:

(i) *nature nodes*, with fixed probability distributions over their successors;

(ii) *prediction nodes*, nature nodes whose distribution may depend on the agent's policy, but only through the actions that the policy prescribes at specified information sets;

(iii) *agent nodes*, partitioned into a finite set H_D of information sets, each $h \in H_D$ carrying a finite nonempty set A_h of actions.

It is required that no information set occurs twice on any path of the tree.

Perfect recall implies the last requirement, which is all that will be used. A *deterministic policy* for D is a choice of action at every information set, and the policy space is

$$\Pi_D = \prod_{h \in H_D} A_h,$$

a finite nonempty set. Mixed policies are deferred to Sections 6 and 13.

The policy-dependent kernels of the prediction nodes determine the distribution of actual play, but they do not by themselves settle the causal counterfactuals that several criteria require: what would have happened had the agent acted otherwise, *with the environment's responses held fixed*. To make these counterfactuals explicit, the tree is supplemented with a structural description of the environment.

Definition 2.2 (Structural representation). A *structural representation* of D consists of

(i) a finite nonempty set S_D of *environment plans*, each of which specifies the response of every nature and prediction node, including nodes on branches not taken and responses that depend on earlier physical actions;

(ii) for every policy $\pi \in \Pi_D$, a probability distribution $\mu_\pi \in \Delta(S_D)$;

(iii) the payoff $U_D(s, \pi)$ obtained by traversing the tree with plan s and policy π.

The *ex-ante value* of π is

$$V_D(\pi) = \sum_{s \in S_D} \mu_\pi(s)\, U_D(s, \pi).$$

The representation separates two kinds of dependence. *Ordinary causal dependence* of the environment on earlier actions is carried by the responses written into each plan s. *Predictive dependence* on the policy is carried by the map $\pi \mapsto \mu_\pi$. A tree with policy-dependent kernels can always be given a structural representation by supplying potential outcomes for its nature and prediction nodes; how the outcomes on alternative branches are coupled is an additional modelling choice, which matters for the causal criteria and is fixed by the representation. Conversely, every structural representation is implemented by a tree whose root is a prediction node selecting s according to μ_π, followed by deterministic traversal.

2.2. Well-definedness and fairness

Two conditions on problems are distinguished.

DEFINITION 2.3 (Well-definedness). D is *well defined* if μ_π is a probability distribution on S_D for *every* $\pi \in \Pi_D$.

DEFINITION 2.4 (Fairness). D is *fair* if S_D, U_D, and the entire map $\pi \mapsto \mu_\pi$ are fixed features of D: the evaluation of an agent may depend on the actions its policy prescribes, but not on the identity, name, or source code of the decision theory that produced the policy. Fixed algorithms may occur as constants of the environment ("the predictor simulates the algorithm g_0"), provided g_0 does not vary with the agent being evaluated.

Fairness excludes the trivial diagonal problem that pays an agent exactly when its action differs from that of a given theory T; such a problem defeats every T and shows nothing about decision theory. Well-definedness is a separate requirement. A verbally described "perfect predictor" may satisfy an action-based fairness condition while its promised behaviour is logically inconsistent for some policy; Section 4.4 gives an example. The presence of a fixed algorithm in the environment does not by itself make a self-referential specification well defined (Remark 7.3).

Write $\mathcal{F}^*$ for the class of well-defined, fair finite problems with a fixed structural representation, closed under the commitment construction of Section 5.

DEFINITION 2.5 (Decision theory). A *decision theory* on a class $\mathcal{F}$ of problems assigns to every $D \in \mathcal{F}$ a policy $T_D \in \Pi_D$. No uniformity, computability, or generality is required: every lookup table is a decision theory.

Totality is part of the definition. A proposed criterion that recommends nothing on some problem therefore fails to define a decision theory on any class containing that problem; it does not define a theory that is silent there.

2.3. Reach and local causal deviation

For a policy π and an information set h, let

$$R_h(\pi) = \{s \in S_D \,:\, h \text{ is reached in the traversal with } (s,\pi)\}, \qquad r_\pi(h) = \mu_\pi\big(R_h(\pi)\big).$$

The number $r_\pi(h)$ is the actual probability that h is reached under π; h is *positively reached* under π if $r_\pi(h) > 0$. For $a \in A_h$, let $\pi[h{\leftarrow}a]$ denote the policy that agrees with π everywhere except at h, where it prescribes a.

LEMMA 2.6. *For every π, h, and $a \in A_h$:*

(i) $R_h(\pi[h{\leftarrow}a]) = R_h(\pi)$;

(ii) if $s \notin R_h(\pi)$, *then* $U_D(s,\pi[h{\leftarrow}a]) = U_D(s,\pi)$.

Proof. With the plan s fixed, the path through the tree is determined by the prescriptions at the information sets it visits. Up to its first visit to h, the path under $(s,\pi[h{\leftarrow}a])$ uses only prescriptions at information sets other than h, which are unchanged; so it coincides with the path under (s,π), and h is reached under one exactly when it is reached under the other. If h is not reached, the two paths coincide entirely. Since no information set recurs on a path, a visit to h is the first and only one. □

The *causal value* of a at a positively reached h, with beliefs induced by π, is

$$Q_\pi(h,a) = \frac{1}{r_\pi(h)} \sum_{s \in R_h(\pi)} \mu_\pi(s)\, U_D\big(s,\pi[h{\leftarrow}a]\big).$$

This fixes the causal counterfactual completely: the distribution over environmental responses is the one the actual policy induces, the action at h is changed, and the policy's other prescriptions are kept. The contrast with ex-ante evaluation lies in a single subscript: $Q_\pi(h,a)$ averages with μ_π, whereas $V_D(\pi[h{\leftarrow}a])$ averages with $\mu_{\pi[h\leftarrow a]}$. Every within-problem conflict studied below is a consequence of that difference.

3. Optimality criteria and the possibility of ex-ante optimality

3.1. Within-problem criteria

The criteria below are the precise forms of the principles that the main decision theories are usually taken to satisfy. Each is a condition on the policy T_D that a theory T selects in a problem D; the commitment criterion C4 and the evidential criterion C6 are introduced in Section 5, once the constructions they require are available.

DEFINITION 3.1 (Ex-ante optimality, C1). T satisfies C1 on D if

$$T_D \in \arg\max_{\pi \in \Pi_D} V_D(\pi).$$

DEFINITION 3.2 (Structural causal dominance, C2). T satisfies C2 on D if, at every information set h positively reached under $\pi = T_D$, there is no $a \in A_h$ such that

$$U_D\big(s,\pi[h{\leftarrow}a]\big) > U_D\big(s,\pi[h{\leftarrow}\pi(h)]\big) \quad \text{for every } s \in R_h(\pi).$$

The quantifier ranges over *structural* states: all plans that reach h, including those to which μ_π assigns probability zero. This is a deliberate resolution of an ambiguity in "every state compatible with h", and it matters. If compatibility meant only the states in the support of μ_π, then in Death in Damascus with an infallible predictor (Section 4.3) the single supported state places Death in the chosen city, so switching would strictly improve the payoff in every supported state and C2 itself would be unsatisfiable. Likewise, on trees where prediction nodes follow h, "hold fixed the outcomes of the nodes preceding h" leaves open how the later prediction nodes are to be evaluated; freezing their policy dependence while averaging over their outcomes can make dominance collapse into an unsatisfiable best-response requirement. The structural plan fixes both matters at once.

Definition 3.3 (Ratifiability, C3). T satisfies C3 on D if, at every information set h positively reached under $\pi = T_D$,

$$\pi(h) \in \arg\max_{a \in A_h} Q_\pi(h, a).$$

Definition 3.4 (Existence, C5). T satisfies C5 on D if its own criterion is satisfiable there, that is, if T_D is defined.

Since decision theories are total by Definition 2.5, C5 holds for every theory and imposes no restriction; it records the requirement that a criterion be satisfiable on every problem of the class. Were partial theories admitted, totality would become an independent axiom and the consistency maps of Section 5 would change.

Proposition 3.5. C3 *implies* C2.

Proof. Suppose h is positively reached under $\pi = T_D$ and some a satisfies $U_D(s, \pi[h \leftarrow a]) > U_D(s, \pi)$ for every $s \in R_h(\pi)$. Since $r_\pi(h) > 0$, the conditional distribution $\mu_\pi(\cdot \mid R_h(\pi))$ is a probability measure carried by $R_h(\pi)$, and a strict pointwise inequality on $R_h(\pi)$ gives $Q_\pi(h, a) > Q_\pi(h, \pi(h))$. Thus $\pi(h)$ is not a maximiser of $Q_\pi(h, \cdot)$, and C3 fails. □

3.2. Ex-ante optimality is always attainable

The opening question admits a direct answer once "decision theory" is taken in the permissive sense of Definition 2.5.

Theorem 3.6 (Universal ex-ante selection). *On every class of well-defined finite problems there is a decision theory satisfying* C1 *on every problem of the class.*

Proof. Fix a tie-breaking order on each Π_D and set $T_D^{\mathrm{ex}} = \text{first} \arg\max_{\pi \in \Pi_D} V_D(\pi)$. Each Π_D is finite and nonempty, and V_D is real valued on it because D is well defined, so the maximum exists. □

The unqualified thesis that no decision theory is optimal on all fair problems is therefore false. What can be true is that particular *combinations* of criteria are

jointly unsatisfiable, and the remaining sections determine exactly which. A first observation shows how little such a result can depend on generality or uniformity. Call a criterion C *local* if whether a theory T satisfies C on D depends only on T_D; C1, C2, C3, and C5 are local, while C4 is not. For a local criterion C_i let $A_i(D) \subseteq \Pi_D$ be the set of policies that satisfy it on D.

LEMMA 3.7 (Pointwise reduction). *For local criteria $C_1, \dots, C_k$ and a class $\mathcal{F}$,*

$$\exists T\ \forall D \in \mathcal{F}\ \bigwedge_i C_i(T, D) \quad \Longleftrightarrow \quad \forall D \in \mathcal{F}\ \bigcap_i A_i(D) \neq \emptyset.$$

Proof. From left to right, T_D lies in the intersection. From right to left, let T_D be the first element of the intersection under a fixed order. □

Because lookup tables are admitted, an impossibility for local criteria is never more than the exhibition of a single problem on which their acceptable sets are disjoint. No difficulty about general rules, computability, or self-reference is involved. This is the precise sense in which the within-problem impossibilities of the next two sections are shallow; non-local criteria, and the cross-problem axioms of Sections 7–11, are where deeper structure can appear.

4. The witness problems

This section works through the problems that separate the criteria. Payoffs are scaled for arithmetic convenience; nothing depends on the particular numbers beyond the inequalities stated.

4.1. Newcomb's problem

Let O denote taking only the opaque box and B taking both boxes. A predictor of symmetric error $\varepsilon \in [0, 1]$ forecasts the agent's prescription at its single information set, which is reached with probability one; the opaque box is full if and only if the forecast is O. Payoffs are

	opaque box full	opaque box empty
O	2	0
B	3	1

with $\Pr(\text{full} \mid O) = 1 - \varepsilon$ and $\Pr(\text{full} \mid B) = \varepsilon$. The structural states are the two contents of the opaque box.

PROPOSITION 4.1 (Nozick 1969). *In Newcomb's problem $V(O) = 2(1 - \varepsilon)$ and $V(B) = 1 + 2\varepsilon$, so $V(O) - V(B) = 1 - 4\varepsilon$ and* C1 *uniquely requires O whenever $0 \leq \varepsilon < \frac{1}{4}$. Since $U(s, B) - U(s, O) = 1$ in both structural states,* C2 *and* C3 *uniquely require B for every ε. Hence no theory satisfies* C1 *and* C2 *on any class containing Newcomb's problem with $\varepsilon < \frac{1}{4}$.*

With opaque prize m and transparent bonus c, the same conflict holds exactly when $m(1-2\varepsilon) > c$. Any symmetric predictor better than chance therefore produces it for suitable stakes. The conflict is Nozick's original one between maximising expected utility and dominance; the content of Proposition 4.1 is only that the formal criteria reproduce it.

The proposition shows that a conflict *can* arise. The next result characterises exactly when it can, and shows that neither accurate prediction nor large stakes is needed.

THEOREM 4.2 (Policy-dependent state laws characterise the conflict). *Let A be a finite set of actions with $|A| \geq 2$, S a finite set of states, and $\mu_a \in \Delta(S)$ the state law induced by prescribing a at a single always-reached choice. The following are equivalent:*

(i) for every payoff function $u : A \times S \to [0,1]$, some ex-ante maximising action is not strictly dominated state by state;

(ii) the laws μ_a, $a \in A$, are all equal.

Proof. (ii)⇒(i). Under a common law μ, if a strictly dominates b in every state then $\mathbb{E}_\mu u(a,\cdot) > \mathbb{E}_\mu u(b,\cdot)$, so no ex-ante maximiser is strictly dominated; a maximiser exists because A is finite.

(i)⇒(ii). Suppose $\mu_a \neq \mu_b$. There is an event $E \subseteq S$ on which they differ; exchanging the names of a and b if necessary, $\delta = \mu_b(E) - \mu_a(E) > 0$. Choose $0 < c < \delta/(1+\delta)$ and set

$$u(b,s) = (1-c)\,\mathbf{1}_E(s), \qquad u(a,s) = (1-c)\,\mathbf{1}_E(s) + c,$$

with payoff 0 for every other action. All payoffs lie in $[0,1]$, and a strictly dominates b by c in every state. Yet

$$V(b) - V(a) = (1-c)\mu_b(E) - (1-c)\mu_a(E) - c = (1-c)\delta - c > 0,$$

and $V(a) \geq c > 0$ exceeds the value 0 of every other action. So b is the unique ex-ante maximiser, and it is strictly dominated. □

Any non-constant dependence of the preceding state law on the prescription thus suffices for some bounded payoff assignment to separate C1 from C2; for fixed payoffs, the size of the dependence determines whether the conflict occurs.

4.2. The twin prisoner's dilemma

Two copies of the same policy play a prisoner's dilemma with temptation 3, reward 2, punishment 1, and sucker's payoff 0. The twin is represented as a prediction node that copies the agent's prescription at its single information set; it is *not* a second, independently assignable coordinate of Π_D, which would remove the stipulated identity of the two decisions. Then $V(\text{cooperate}) = 2$ and $V(\text{defect}) = 1$, so C1 requires cooperation, while with the twin's action held fixed, defection gains one unit whether the twin cooperates or defects, so C2 and C3 require defection. The connection between this problem and Newcomb's is due to Lewis (1979). What the tree

does not supply is the identity of the two decisions: "the same computation" enters only as a stipulated dependence of the prediction node on the agent's prescription.

4.3. Death in Damascus

Death predicts which of two cities the agent will choose and waits there (Gibbard and Harper 1978). Let actions and states both be $\{1, 2\}$, with $U(s, a) = \mathbf{1}\{a \neq s\}$ and $\mu_a = \delta_a$ (infallible prediction).

PROPOSITION 4.3. *In Death in Damascus no deterministic policy satisfies* C3*, whereas both deterministic policies satisfy* C2*. The failure of* C3 *persists for every symmetric error* $\varepsilon < \frac{1}{2}$.

Proof. Under the policy a, the induced state is a; staying has causal value 0 and switching has causal value 1. With error ε, staying has causal value ε and switching $1 - \varepsilon$, and $\varepsilon < 1 - \varepsilon$. For C2, neither city is better than the other in *both* structural states, so no action is dominated. □

Consequently C3, read as a requirement on deterministic policies, is individually unsatisfiable on any class containing this problem: it fails to define a decision theory at all. The absence of ratifiable pure acts in this problem is well known from the ratifiability literature (Skyrms 1990a); Section 6.3 records how far randomisation repairs it.

4.4. The Bomb

MacAskill (2019) describes a predictor of error 10^{-24} that places a bomb in the left box if and only if it predicts that the agent will take the right box, and leaves a note saying so. The agent sees the note and the bomb. Taking the left box is free; taking the right box costs 1; burning to death costs $M = 10^9$ in the scaling used here.

Once the agent observes whether a bomb is present, there are *two* information sets, h_{bomb} and h_{clear}, and four policies $(a_{\text{bomb}}, a_{\text{clear}}) \in \{L, R\}^2$. This has a consequence for well-definedness.

PROPOSITION 4.4 (The Bomb as stated is not well defined). *No predictor whose forecast determines the observation ("predicts R: bomb; predicts L: no bomb") and whose target is the agent's* eventual *action can have an error bound below one uniformly over the four policies.*

Proof. Under the policy (L, R), a forecast of R produces a bomb and hence the action L; a forecast of L produces no bomb and hence the action R. The forecast is wrong in every case. □

A coherent version is obtained by letting the predictor target the fixed coordinate $\pi(h_{\text{bomb}})$, whether or not h_{bomb} is eventually reached:

$$\Pr\big(\text{bomb} \mid a_{\text{bomb}} = L\big) = \varepsilon, \qquad \Pr\big(\text{bomb} \mid a_{\text{bomb}} = R\big) = 1 - \varepsilon.$$

PROPOSITION 4.5. *In the repaired Bomb with death cost $M > 1$, the ex-ante values are*

policy $(a_{\text{bomb}}, a_{\text{clear}})$	V
(L, L)	$-M\varepsilon$
(L, R)	$-M\varepsilon - (1-\varepsilon)$
(R, L)	$-(1-\varepsilon)$
(R, R)	-1

For $0 < \varepsilon < 1/(M+1)$, C1 uniquely selects (L, L), while C2 and C3 uniquely select (R, L).

Proof. The values follow by conditioning on the presence of the bomb. (L, L) beats (R, L) exactly when $M\varepsilon < 1 - \varepsilon$, that is $\varepsilon < 1/(M+1)$; it beats (L, R) and (R, R) a fortiori. For $\varepsilon > 0$ both information sets are positively reached under every policy. At h_{bomb}, every structural state reaching it contains the bomb, and R yields $-1 > -M$; at h_{clear}, L yields $0 > -1$. Hence (R, L) is the only policy satisfying C2, and, since it is causally optimal at both information sets, it satisfies C3. □

MacAskill's numbers satisfy the inequality. Two qualifications are essential. At $\varepsilon = 0$, h_{bomb} is unreached under (L, L), and the reached-only criteria C2 and C3 do not oppose that policy. And if the problem is instead taken to be the already-conditioned decision "a bomb is visibly present", its two options have values $-M$ and -1, and C1 itself selects R. The visible-bomb challenge is MacAskill's; the complete contingent-policy kernel above is the additional assumption under which it becomes a theorem.

4.5. Counterfactual mugging

A fair coin is tossed. On heads the agent receives r if and only if an infallible predictor forecasts that the agent's prescription on tails is to pay; on tails the agent is asked to pay c, with $r > c > 0$ (Nesov 2009). There is one information set, reached on tails. Then

$$V(\text{Pay}) = \tfrac{1}{2}(r - c) > 0 = V(\text{Refuse}),$$

so C1 requires paying, while at the tails information set paying costs c in every structural state reaching it, so C2 and C3 require refusing. The problem separates ex-ante from ex-post evaluation by a prediction about an information set that is unreached on half the paths; it is used in Section 5.3 to locate evidential criteria and in Section 6.4 to test restrictions on what predictors may target.

5. COMMITMENT, EVIDENCE, AND THE WITHIN-PROBLEM MAP

5.1. The commitment extension

Reflective stability asks what a theory would do if, before any prediction is made, it could choose which policy it will run. To state this as a criterion, the choice of a policy must itself be a decision problem.

DEFINITION 5.1 (Commitment extension). Let D be a problem and λ a *coupling* of the laws $(\mu_q)_{q\in\Pi_D}$, that is, a probability distribution on $\prod_{q\in\Pi_D}\operatorname{supp}\mu_q$ whose marginal on the q-th coordinate is μ_q. The *commitment extension* $K_\lambda(D)$ is the problem in which

(i) the agent, at a single information set, chooses a policy $q \in \Pi_D$;

(ii) this choice irrevocably installs q;

(iii) the environment draws $\omega = (s_q)_{q\in\Pi_D}$ from λ, independently of the choice, and the agent receives $U_K(\omega, q) = U_D(s_q, q)$.

Unless stated otherwise the structural states of $K_\lambda(D)$ are $S_K = \prod_q \operatorname{supp}\mu_q$, and $K(D)$ denotes the product coupling $\lambda = \bigotimes_q \mu_q$. The constructor is tagged and well founded, so that $K(D)$, $K(K(D))$, … have unambiguous decompositions, and $\mathcal{F}^*$ is closed under it.

The extension models a *causal* choice of policy made before prediction: changing the commitment changes the environment's subsequent response, because the coordinate s_q that is paid out depends on q, while the law λ of the whole vector does not.

PROPOSITION 5.2. *For every coupling λ and every $q \in \Pi_D$, $V_{K_\lambda(D)}(q) = V_D(q)$.*

Proof. $V_{K_\lambda(D)}(q) = \mathbb{E}_\lambda\big[U_D(s_q, q)\big] = \mathbb{E}_{\mu_q}\big[U_D(s, q)\big] = V_D(q)$, since the q-th marginal of λ is μ_q. □

DEFINITION 5.3 (Commitment stability, C4). T satisfies C4 on D if the commitment $q = T_{K(D)}$ that T chooses in $K(D)$ satisfies $q(h) = T_D(h)$ at every information set h positively reached under T_D.

Unlike the other criteria, C4 relates a theory's behaviour on two different problems. It is the first cross-problem condition in this paper, and the only one stated without passing to the reduced form of Section 7.

5.2. The map of C1–C5

The consistent combinations need explicit total witnesses, not merely examples on a few problems.

PROPOSITION 5.4 (A coherent ex-ante theory). *There is a decision theory on $\mathcal{F}^*$ satisfying* C1 *and* C4.

Proof. On problems that are not commitment extensions, let $T_D = T_D^{\mathrm{ex}}$. Recursively, in $K(D)$ let T commit to the policy T_D already selected for D. By Proposition 5.2, T_D maximises $V_{K(D)}$, so C1 holds on $K(D)$; C4 holds with exact agreement. □

C1 alone does not imply C4: on a problem in which both actions yield zero, an ex-ante maximising lookup table may choose different tied actions in D and in $K(D)$.

PROPOSITION 5.5 (A dominance-respecting theory). *For each D fix a full-support distribution ν_D on S_D, for instance the uniform one, and let*

$$T_D^{\nu} = \operatorname{first}\arg\max_{\pi\in\Pi_D} \sum_{s\in S_D} \nu_D(s)\,U_D(s,\pi).$$

Then T^{ν} satisfies C2 *on every problem.*

Proof. Let $\pi = T_D^{\nu}$ and suppose that at an information set h positively reached under π some a satisfies $U_D(s, \pi[h{\leftarrow}a]) > U_D(s,\pi)$ for all $s \in R_h(\pi)$. By Lemma 2.6(ii) the payoff is unchanged on $S_D \setminus R_h(\pi)$. The set $R_h(\pi)$ is nonempty because $r_\pi(h) > 0$, and ν_D gives it positive weight, so $\pi[h{\leftarrow}a]$ strictly improves the objective that π maximises, a contradiction. □

T^{ν} is an existence witness, not a recommended theory: its fixed weights ignore the predictive dependence altogether.

PROPOSITION 5.6. *No theory satisfies* C2 *and* C4 *on a class containing Newcomb's problem with $\varepsilon = 0$ and its commitment extension.*

Proof. In D, C2 requires B (Proposition 4.1). At $\varepsilon = 0$ every support is a singleton, so $K(D)$ has the single structural state (opaque box full under O, empty under B), and committing to O yields 2 while committing to B yields 1. Committing to O therefore strictly dominates, and C2 in $K(D)$ requires it; C4 requires the commitment to agree with B at the always-reached information set. □

THEOREM 5.7 (Within-problem map). *On $\mathcal{F}^*$ the inclusion-minimal inconsistent subsets of* $\{\mathrm{C1},\dots,\mathrm{C5}\}$ *are exactly*

$$\{\mathrm{C3}\},\qquad \{\mathrm{C1},\mathrm{C2}\},\qquad \{\mathrm{C2},\mathrm{C4}\},$$

and the maximal consistent subsets are exactly $\{\mathrm{C1},\mathrm{C4},\mathrm{C5}\}$*, witnessed by Proposition 5.4, and* $\{\mathrm{C2},\mathrm{C5}\}$*, witnessed by Proposition 5.5.*

Proof. The three sets are inconsistent by Propositions 4.3, 4.1, and 5.6; their proper subsets are consistent by the two witnesses (C5 holds for every theory). A consistent set cannot contain C3. If it contains C2, it contains neither C1 nor C4, so it lies in $\{\mathrm{C2},\mathrm{C5}\}$. Otherwise its members lie in $\{\mathrm{C1},\mathrm{C4},\mathrm{C5}\}$. □

Among C1–C5, then, there is no minimally inconsistent family of three or more individually satisfiable criteria. The requested minimality is attained only by the pairs $\{\mathrm{C1},\mathrm{C2}\}$ and $\{\mathrm{C2},\mathrm{C4}\}$, whose singletons have explicit total witnesses, and the first is a direct pointwise disagreement in the sense of Lemma 3.7.

5.3. An evidential criterion

An evidential criterion requires a joint law of acts and states. Under a deterministic policy, conditioning on an alternative action is ordinarily conditioning on a null

event, so the expression $\mathbb{E}[\text{payoff} \mid h, \text{agent takes } a]$ cannot be read off the actual distribution of play. The following completion makes it determinate.

Definition 5.8 (Reference-policy evidential criterion, C6$^\kappa$). For each D fix a full-support distribution κ_D on Π_D and define the joint law $J_D(\Pi = \pi, S = s) = \kappa_D(\pi)\mu_\pi(s)$, with $Y = U_D(S, \Pi)$. For $h \in H_D$ and $a \in A_h$ let

$$e_h(a) = \mathbb{E}_{J_D}\big[Y \mid h \text{ reached},\ \Pi(h) = a\big],$$

with $e_h(a) = -\infty$ when the conditioning event is null. T satisfies C6$^\kappa$ on D if $T_D(h) \in \arg\max_a e_h(a)$ at every h positively reached under T_D.

This is a specified evidential criterion, not an identification with every version of evidential decision theory; in sequential problems it evaluates continuation behaviour with the reference distribution over policies. It is individually satisfiable: choose an e_h-maximiser at every h with $J_D(h \text{ reached}) > 0$ and anything elsewhere. If h is positively reached under T_D, then $J_D(h \text{ reached}) \geq \kappa_D(T_D)\, r_{T_D}(h) > 0$, so the constraint is met, and some action has a finite evidential value there.

Proposition 5.9. *(i) In Newcomb's problem $e(O) = V(O)$ and $e(B) = V(B)$, so C6$^\kappa$ requires O for $\varepsilon < \frac{1}{4}$ and conflicts with C2.*

(ii) In counterfactual mugging C6$^\kappa$ requires refusing, since $e(\mathit{Pay}) = -c < 0 = e(\mathit{Refuse})$ at the tails information set, and conflicts with C1.

(iii) In the commitment extension of counterfactual mugging, $e(q) = V_D(q)$, so C6$^\kappa$ commits to paying while refusing in the original problem, and conflicts with C4.

Proof. (i) With one always-reached information set, conditioning on the action identifies the policy, so $e(a) = \mathbb{E}_{\mu_a} U(\cdot, a) = V(a)$. (ii) Conditional on tails and on the tails prescription being "pay", the payoff is $-c$; on "refuse" it is 0. (iii) In $K(D)$ the commitment is the policy variable and λ does not depend on it, so $e(q) = \mathbb{E}_\lambda U_D(s_q, q) = V_D(q)$, and $V_D(\text{Pay}) > V_D(\text{Refuse})$; the tails information set is reached with probability $\frac{1}{2}$ under the refusing policy. □

Theorem 5.10 (Map with the evidential criterion). *On $\mathcal{F}^*$ the inclusion-minimal inconsistent subsets of $\{\text{C1}, \dots, \text{C5}, \text{C6}^\kappa\}$ are*

$$\{\text{C3}\},\quad \{\text{C1}, \text{C2}\},\quad \{\text{C2}, \text{C4}\},\quad \{\text{C1}, \text{C6}^\kappa\},\quad \{\text{C2}, \text{C6}^\kappa\},\quad \{\text{C4}, \text{C6}^\kappa\},$$

and the maximal consistent subsets are $\{\text{C1}, \text{C4}, \text{C5}\}$, $\{\text{C2}, \text{C5}\}$, and $\{\text{C6}^\kappa, \text{C5}\}$.

Proof. Among C1, C2, C4, C6$^\kappa$, every pair except $\{\text{C1}, \text{C4}\}$ is inconsistent by Propositions 4.1, 5.6, and 5.9, and each listed maximal set has a total witness. □

A different evidential semantics can change this table; Section 13.3 compares two. The theorem is a statement about the displayed definitions.

5.4. *Noisy commitment and couplings*

Proposition 5.6 uses exact prediction and, implicitly, a choice of coupling. Both matter, and they matter differently for structural and for expectational comparisons. Let the opaque prize be m and the transparent bonus c, with $m > c > 0$, and let the symmetric error be $\varepsilon \in [0, 1]$. Write $g = m(1 - 2\varepsilon) - c$ for the ex-ante advantage of O.

Let X indicate a full opaque box under commitment to O, and Y a full opaque box under commitment to B, so that $\Pr(X = 1) = 1 - \varepsilon$ and $\Pr(Y = 1) = \varepsilon$. Every coupling is determined by $t = \Pr(X = 1, Y = 1)$, which ranges over $0 \le t \le \min(\varepsilon, 1 - \varepsilon)$:

(X, Y)	probability	payoff of O	payoff of B	difference $O - B$
$(1, 1)$	t	m	$m + c$	$-c$
$(1, 0)$	$1 - \varepsilon - t$	m	c	$m - c$
$(0, 1)$	$\varepsilon - t$	0	$m + c$	$-m - c$
$(0, 0)$	t	0	c	$-c$

PROPOSITION 5.11 (Noisy commitment in Newcomb's problem). *(i) For every coupling, the expected causal value of committing to q is $V_D(q)$ whatever the agent's beliefs about its own commitment; every criterion that ranks commitments by expected causal value requires commitment to O exactly when $g > 0$, while the same criterion requires B in D.*

(ii) With structural states $S_K = \operatorname{supp} \lambda$, commitment to O strictly dominates commitment to B if and only if $\varepsilon = 0$; commitment to B strictly dominates if and only if $\varepsilon \ge \frac{1}{2}$ and $t = 1 - \varepsilon$.

(iii) With structural states $S_K = \prod_q \operatorname{supp} \mu_q$, neither commitment dominates for $0 < \varepsilon < 1$, whatever the coupling.

Consequently the C2–C4 *conflict of Proposition 5.6 occurs, on this pair of problems, only at $\varepsilon = 0$ and under every coupling, while the reversal of expectational comparisons between D and $K_\lambda(D)$ occurs whenever $g > 0$, independently of λ.*

Proof. (i) Since λ does not depend on the commitment, the causal and the ex-ante value of committing to q coincide and equal $V_D(q)$ by Proposition 5.2. In D the bonus c is gained in every state, so every expectational causal comparison favours B. (ii) O dominates only if the three rows with negative difference are null: $t = 0$ and $\varepsilon - t = 0$, so $\varepsilon = 0$; conversely, at $\varepsilon = 0$ only the row $(1, 0)$ remains. Directly: at positive error O pays 0 with positive probability whereas B always pays at least $c > 0$, so no coupling makes O even weakly dominant. B dominates exactly when the row $(1, 0)$ is null, $t = 1 - \varepsilon$, which the constraint $t \le \min(\varepsilon, 1 - \varepsilon)$ allows only for $\varepsilon \ge \frac{1}{2}$. (iii) For $0 < \varepsilon < 1$ all four rows are structural states, and the differences have both signs. □

Changing a probability law cannot change C2 while its structural state set is held fixed; the coupling matters for C2 only through the convention $S_K = \operatorname{supp} \lambda$. The three couplings one might call natural give

coupling	t
independent (product)	$\varepsilon(1-\varepsilon)$
shared predictor-correctness bit	0
comonotone realised payoffs	$\min(\varepsilon, 1-\varepsilon)$

and none of them restores dominance of O at positive error. Couplings do matter in other problems: for a shared Bernoulli variable Z, the payoffs $U = 1 + 100Z$ and $W = 100Z$ satisfy $U > W$ everywhere on the support of the shared coupling, whereas the product coupling admits the reversing pair $U = 1$, $W = 100$. Which couplings produce dominance is settled in general by the following criterion, a finite form of Strassen's theorem on stochastic order (Strassen 1965).

PROPOSITION 5.12 (Dominance under some coupling). *Let ν_U, ν_W be finitely supported laws on $\mathbb{R}$ with distribution functions F_U, F_W.*

(i) There is a coupling with $U \geq W$ almost surely if and only if $F_U(x) \leq F_W(x)$ for all x.

(ii) There is a coupling with $U > W$ almost surely if and only if $F_U(x) \leq F_W(x^-)$ for every $x \in \operatorname{supp} \nu_U$.

Proof. Necessity follows from the inclusions $\{U \leq x\} \subseteq \{W \leq x\}$ and $\{U \leq x\} \subseteq \{W < x\}$, which hold almost surely under the respective couplings. For sufficiency, let V be uniform on $(0, 1]$ and set $U = F_U^{-1}(V)$, $W = F_W^{-1}(V)$, with $F^{-1}(v) = \min\{y : F(y) \geq v\}$. In case (i), $F_U \leq F_W$ gives $F_W^{-1} \leq F_U^{-1}$ pointwise. In case (ii), let $x = F_U^{-1}(v) \in \operatorname{supp} \nu_U$; then $F_W(x^-) \geq F_U(x) \geq v$, and since ν_W is finitely supported, $F_W(y) \geq v$ for some $y < x$, so $F_W^{-1}(v) < x$. □

A payoff coupling lifts to a coupling of the underlying states by distributing mass within payoff fibres according to the conditional state laws. For a general problem, the residual content of C4 is then an agreement condition: some C2-permitted original policy must agree, on the information sets it positively reaches, with a commitment that no other commitment structurally dominates. Proposition 5.12 decides which commitments are dominated; it does not by itself decide whether C2 and C4 are consistent on a whole class of problems with imperfect prediction, which remains open (Section 14).

6. Robustness of the within-problem conflict

The central conflict between C1 and C2 was obtained with deterministic policies, a single always-reached choice, and a predictor targeting that choice. This section asks which of these features it needs.

6.1. Imperfect prediction

By Proposition 4.1, the conflict holds for every $0 \leq \varepsilon < \frac{1}{4}$ at the scaled payoffs, and for every $\varepsilon < \frac{1}{2}$ at suitable stakes, since $m(1-2\varepsilon) > c$ can be arranged whenever $\varepsilon < \frac{1}{2}$.

At $\varepsilon = \frac{1}{2}$ the symmetric predictor carries no policy dependence and this mechanism disappears. Theorem 4.2 gives the general statement: what is needed is not accuracy but any difference between the state laws that two prescriptions induce.

6.2. Randomisation does not reconcile ex-ante and causal evaluation

Let the agent one-box with probability p. Suppose first that the predictor observes the distribution, filling the box with probability $\varepsilon + (1 - 2\varepsilon)p$, and that the action is drawn independently given p. Expanding,

$$V(p) = p \cdot 2\big(\varepsilon + (1-2\varepsilon)p\big) + (1-p)\big(2\varepsilon + 2(1-2\varepsilon)p + 1\big) = 1 + 2\varepsilon + (1-4\varepsilon)p.$$

The same formula holds if the predictor instead observes the sampled action and predicts it with symmetric error ε. For $\varepsilon < \frac{1}{4}$, C1 requires $p = 1$ and C2 requires $p = 0$. The conflict also admits an exact quantitative form. Put $g = 1 - 4\varepsilon > 0$, let $R_1(p) = \max_q V(q) - V(p)$ be ex-ante regret, and let $R_{\text{causal}}(p)$ be the expected regret relative to the best action in the realised state.

PROPOSITION 6.1 (Exact trade-off). *In randomised Newcomb, $R_1(p) = g(1-p)$ and $R_{\text{causal}}(p) = p$, so that*

$$\frac{R_1(p)}{g} + R_{\text{causal}}(p) = 1.$$

Simultaneous tolerances $R_1 \le \delta_1$ and $R_{\text{causal}} \le \delta_2$ are achievable if and only if $\delta_1/g + \delta_2 \ge 1$. Under infallible prediction, every policy has regret at least $\frac{1}{2}$ by one of the two measures.

Proof. V is affine with slope g and maximal at $p = 1$, which gives R_1. In every realised state two-boxing is better by exactly one unit, so the causal regret is 1 when the agent one-boxes and 0 otherwise. The tolerances require $1 - \delta_1/g \le p \le \delta_2$, a nonempty interval exactly under the stated inequality. At $\varepsilon = 0$, $g = 1$ and $\max(R_1, R_{\text{causal}}) \ge \frac{1}{2}$. □

6.3. Existence of optimal and ratifiable mixed policies

Admitting mixed policies does not by itself make either C1 or C3 satisfiable; regularity of the predictor's dependence on the distribution is what matters.

PROPOSITION 6.2. *(i) Let a binary reward, independent of the realised action, succeed with probability $q(p) = p$ for $p < 1$ and $q(1) = 0$. Then $V(p) = q(p)$ has supremum 1 and no maximiser.*

(ii) Let Death choose city 1 if and only if $p \ge \frac{1}{2}$, where p is the probability of city 1, with payoff 1 for avoiding Death. No mixture is ratifiable, although Death observes only the distribution.

Proof. (i) is immediate. (ii) If $p < \frac{1}{2}$, Death is in city 0, and city 1 is the unique causal best response, so a ratifiable p would put all its mass on city 1; if $p \ge \frac{1}{2}$, city 0 is the unique best response. Either way p is not supported on best responses to itself. □

THEOREM 6.3 (Existence under continuity). *Let actions and states be finite, let the action be sampled independently given the mixed policy $p \in \Delta(A)$, and let $p \mapsto \mu_p \in \Delta(S)$ be continuous. Then a mixed policy satisfying* C1 *exists, and so does a mixed policy all of whose supported actions are causal best responses to μ_p.*

Proof. $V(p) = \sum_a p_a \sum_s \mu_p(s)u(s,a)$ is continuous on the compact simplex, so it attains its maximum. For ratifiability, let $Q(a;p) = \sum_s \mu_p(s)u(s,a)$ and $B(p) = \Delta\big(\arg\max_a Q(a;p)\big)$. The correspondence B has nonempty, compact, convex values, and its graph is closed because the finitely many functions $Q(a;\cdot)$ are continuous. By Kakutani's theorem (Kakutani 1941) there is $p \in B(p)$. □

Theorem 6.3 is the one-shot core of two published existence results with stronger structure: Halpern's theorem on ratifiable acts, which assumes costless statewise mixing and conditional state laws that are affine in the mixture (Halpern 2020), and the theorem of Bell et al. (2021) that every continuous Newcomblike decision process has a strongly ratifiable policy. Neither identifies ratifiable policies with ex-ante maximisers, and Proposition 6.1 shows that they cannot be identified.

REMARK 6.4 (Predicting a realised sample). If Death independently samples the agent's distribution, the usual symmetric model admits $p = \frac{1}{2}$. If Death instead predicts the agent's realised private sample, the verdict depends on further semantics: whether the agent knows its sample, and whether a deviation preserves or replaces the randomness the predictor tracks. Conditioning on a revealed, perfectly predicted sample restores the deterministic instability. A marginal test of support-optimality can still accept $p = \frac{1}{2}$ while ignoring the correlation between the prediction and the sample, and should not be confused with a criterion that preserves that correlation when the unchanged policy is evaluated. Skyrms already cautioned against an unconditional randomisation rescue (Skyrms 1990a).

6.4. Restrictions on what predictors may target

Some responses to Newcomb-type problems restrict prediction to information sets that are in some sense actual. Newcomb's problem has one information set, reached with probability one under every policy, so its conflict survives every such restriction except the last:

admissible prediction targets	Newcomb conflict survives
information sets positively reached under the evaluated policy	yes
information sets reached under every admissible policy	yes
information sets on the realised path	yes
information sets already reached before the prediction occurs	no

The last restriction removes anticipatory dependence altogether: a "prediction" of an action already taken is an ordinary causal response, which belongs in the structural plan, so μ_π no longer depends on π. In that case ex-ante and causal evaluation agree.

PROPOSITION 6.5. *If $\mu_\pi = \mu$ does not depend on π, then every* C1*-optimal policy satisfies* C3*, and therefore* C2*.*

Proof. By Lemma 2.6,

$$V_D(\pi[h{\leftarrow}a]) - V_D(\pi) = \sum_{s \in R_h(\pi)} \mu(s)\big(U_D(s, \pi[h{\leftarrow}a]) - U_D(s,\pi)\big)$$
$$= r_\pi(h)\big(Q_\pi(h,a) - Q_\pi(h,\pi(h))\big).$$

If $r_\pi(h) > 0$ and $\pi(h)$ is not causally optimal at h, the deviation raises the ex-ante value. □

Removing predictions about unreached information sets eliminates particular witnesses, such as counterfactual mugging, and so changes the evidential part of the map in Theorem 5.10. It leaves the central Newcomb incompatibility intact.

7. The causal matrix and cross-problem axioms

Theorem 5.7 shows that within-problem criteria yield only singleton and pairwise obstructions. If a non-trivial impossibility in the style of Arrow exists, it must therefore come from requirements that relate a theory's choices on *different* problems: invariance, independence, contraction. Such requirements need a common description of problems, which the following reduced form provides.

7.1. Reduced form

DEFINITION 7.1 (Causal matrix). The *causal matrix* of a problem D is the square matrix indexed by Π_D,

$$M_D[\pi,\sigma] = \sum_{s \in S_D} \mu_\pi(s)\, U_D(s,\sigma),$$

the value of running σ while the environment responds as it would to π.

Row π is the causal evaluation of every policy under the beliefs that π induces; column σ records how σ fares under every such belief state; and the diagonal is the ex-ante value, $M_D[\pi,\pi] = V_D(\pi)$. The matrix compresses the problem: it retains expectations under each induced law, and it forgets the structural states. Two consequences should be kept in view. The column-dominance axiom below compares expected values, whereas C2 compares structural outcomes, including states of probability zero; and "π maximises its own row" is a global form of ratifiability, which differs from the local, information-set form C3 in multi-stage problems. For problems with a single choice, the matrix and the structural criteria agree, and all the witnesses below are of that kind.

THEOREM 7.2 (Realisability). *Every finite real square matrix is the causal matrix of a fair, well-defined problem with a single information set. Deleting a set of indices together with their rows and columns is realised by deleting the corresponding actions while keeping the remaining laws.*

Proof. Given $M \in \mathbb{R}^{I\times I}$, let the agent choose $i \in I$ at a single information set; let the states be $S = I$; let an infallible predictor forecast the chosen index, $\mu_i = \delta_i$; and let $U(s, j) = M_{sj}$. Then $M_D[i, j] = U(i, j) = M_{ij}$. The problem is fair and well defined, and deleting actions leaves the laws of the others unchanged. □

In this realisation C1 selects the maximal diagonal entries, and C3 accepts exactly the indices i whose diagonal entry is maximal in row i. Rules on matrices therefore have an unrestricted domain, as social welfare functions do in Arrow's theorem (Arrow 1951), and the framework contains every finite recommendation structure.

REMARK 7.3 (Self-referential problems). Skalse (2021) proposes, for a fixed target theory X, a choice between a transparent box containing 1000 and an opaque box whose content is 0 if X permits choosing it in that problem and 2000 otherwise, and argues that another theory can outperform X. The target theory is an environmental constant, so this is not discrimination by the identity of the evaluated agent. Translated into the present model, however, the construction requires a fixed point: for a theory that maximises the diagonal, an empty opaque box makes it choose the transparent box, which requires the opaque box to be full, and conversely. Neither content yields a well-defined instance. In the same way, the diagonal family D_b with $u(a) = \mathbf{1}\{a \neq b\}$, $b \in \{0, 1\}$, would need $b = T(D_b)$, but an ex-ante optimiser has $T(D_0) = 1$ and $T(D_1) = 0$. The schema does not refute Theorem 3.6 on well-defined problems; its additional self-referential realisability assumption is where it leaves that class. Lewis (1981a) analysed a related failure: an announced reward for supposedly irrational choice can render the proposed scenario inconsistent.

7.2. Rules and axioms

Fix an infinite universe $\mathbb{L}$ of labels. A *matrix* is a pair (I, M) with $I \subseteq \mathbb{L}$ finite and nonempty and $M \in \mathbb{R}^{I\times I}$; for $\emptyset \neq K \subseteq I$, $M|_K$ is the *principal restriction* to $K \times K$, which keeps labels. A bijection $\sigma : I \to I'$ acts by $(\sigma \cdot M)_{\sigma(i)\sigma(j)} = M_{ij}$.

DEFINITION 7.4 (Rule). A *rule* assigns to every matrix a reflexive relation $\succeq_M$ on I, with strict part $\succ_M$, and recommends its maximal elements,

$$G(M) = \{i \in I : \text{there is no } j \in I \text{ with } j \succ_M i\},$$

which are required to be nonempty for every M.

The convention makes the comparison axioms below constrain the same relation that determines choice, and it prevents a rule that lacks an ordering from recommending nothing. A decision theory satisfies *reduction* (A0) if its choice on D depends

only on the labelled matrix M_D, so that it selects from $G(M_D)$ for some rule. A *single-valued* deterministic selector cannot be neutral: on the zero 2×2 matrix, swapping the indices fixes the input but moves either possible answer. Choice correspondences avoid this separate obstruction.

Definition 7.5 (Cross-problem axioms). A rule may satisfy the following conditions, for all matrices M and all indices i, j.

N *Neutrality.* For every bijection $\sigma : I \to I'$: $i \succeq_M j \iff \sigma(i) \succeq_{\sigma \cdot M} \sigma(j)$.

F *Common affine invariance.* $\succeq_{aM+b\mathbf{1}\mathbf{1}^\top} = \succeq_M$ for $a > 0, b \in \mathbb{R}$.

R *Row translations.* $\succeq_{M+b\mathbf{1}^\top} = \succeq_M$ for every $b \in \mathbb{R}^I$ (adding b_k to every entry of row k).

P *Column Pareto.* If $M_{k\sigma} \geq M_{k\tau}$ for every $k \in I$, strictly for some k, then $\sigma \succ_M \tau$.

L *Restricted Pareto.* The same, with k ranging only over $\{\sigma, \tau\}$.

D *Diagonal optimality.* $G(M) \subseteq \arg\max_i M_{ii}$.

J *Global ratifiability.* Every $i \in G(M)$ satisfies $M_{ii} \geq M_{ij}$ for all j.

B *Binary independence.* $i \succeq_M j \iff i \succeq_{M|_{\{i,j\}}} j$.

S *Self-sovereignty.* If $M, M' \in \mathbb{R}^{I \times I}$ have the same rows i and j, then $i \succeq_M j \iff i \succeq_{M'} j$.

C *Contraction.* $G(M) \cap K \subseteq G(M|_K)$ for every nonempty $K \subseteq I$.

O *Ordering.* $\succeq_M$ is complete and transitive.

Q *Non-triviality.* For some I there are $M, M' \in \mathbb{R}^{I \times I}$ with $G(M) \neq G(M')$.

P is the expectational counterpart of dominance; D is C1; J is ratifiability of pure choices; S is the condition of Briggs (2010, §6), under which the comparison of two options depends only on how each of them evaluates all options; C is Chernoff's condition, Sen's property α (Chernoff 1954; Sen 1971); B is an independence of irrelevant alternatives for binary comparisons, of the kind that Podgorski (2022) imposes on overall-value theories, required here across menu sizes. Neutrality is stated for bijections between different label sets, because the proofs below transport choices from one menu to another; invariances are imposed on the relation, not only on its maximal elements. Ordering concerns the actual output relation at each input. Requiring only that each $G(M)$ be rationalisable by some weak order would be vacuous, since any nonempty set can be placed in a top indifference class; requiring one order to rationalise the choices from all restrictions of a given matrix is stronger and already entails contraction (Arrow 1959).

Lemma 7.6. B *implies* S *and* C.

Proof. If rows i and j coincide in M and M', so does the principal block on $\{i, j\}$, and B transfers the comparison. For C, let $i \in G(M) \cap K$. Any $j \in K$ with $j \succ_{M|_K} i$ would satisfy $j \succ_{M|_{\{i,j\}}} i$ and hence $j \succ_M i$, contradicting the maximality of i in M. □

The following monotonicity condition is used in Section 11. Write E_{ii} for the matrix with a single entry 1 in position (i, i).

DEFINITION 7.7 (Diagonal monotonicity, Mon). If $M' = M + \delta E_{ii}$ with $\delta \geq 0$, then $i \in G(M)$ implies $i \in G(M')$, and every true comparison $i \succeq_M j$ or $i \succ_M j$ remains true in M'.

7.3. What row-translation invariance means

The axiom R will turn out to be decisive, and it is tempting to read it as the matrix form of causal evaluation. It deserves a precise interpretation.

THEOREM 7.8 (Factorisation). *A rule satisfies* R *if and only if $\succeq_M$ depends on M only through the row-normalised matrix $Z(M)_{ij} = M_{ij} - M_{ii}$, equivalently through the within-row differences of M.*

Proof. Row translations leave Z unchanged. Conversely, if $Z(M) = Z(M')$, then $M'_{ij} - M_{ij} = M'_{ii} - M_{ii}$ is constant along each row, so M' is a row translation of M. □

On a restricted domain, R is factorisation through the image of Z; the zero-diagonal representative need not itself be admissible. With this, familiar criteria can be classified once a matrix semantics is fixed for each.

criterion	matrix semantics	satisfies R?
fixed-prior causal decision theory	rank by $w^\top M$ for a fixed weight vector w	yes
pure global ratifiability	admit i iff $M_{ii} \geq M_{ij}$ for all j	yes, as an admissibility condition
mixed ratifiability	$p \in \mathrm{BR}(p^\top M)$	yes, as an equilibrium set
deliberational causal decision theory	requires initial beliefs, update dynamics, and a final selection	depends on these
evidential theory with reference-policy law	maximise M_{ii}, with $P(i, s) = \rho_i \mu_i(s)$	no
ex-ante policy maximisation	maximise $M_{ii} = V(i)$	no

Ratifiability's invariance does not establish that ratifiable options exist or supply a ranking. For deliberational theories, fixed-belief causal comparisons with difference-based update dynamics preserve row baselines, but arbitrary update dynamics need not; Joyce (2012) distinguishes equilibrium credences from the act subsequently chosen, and Arntzenius (2008) identifies the update rule as additional structure needed to define equilibrium. An "FDT-style" theory is ex-ante maximisation under stipulated functional counterfactuals and so fails R under that stipulation (Yudkowsky and Soares 2017).

PROPOSITION 7.9. R *does not characterise causal rationality: ranking by* $r_i = M_{ii} - \max_j M_{ij}$ *satisfies* N, F, R, O, *yet on*

$$M = \begin{pmatrix} 0 & 10 & -1 \\ 10 & 0 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

it chooses option 3, *which both other columns strictly dominate.*

Proof. The scores are $(-10, -10, -1)$. Column 3 is $(-1, -1, -1)^\top$, below both $(0, 10, 0)^\top$ and $(10, 0, 0)^\top$ in every row. □

So R means exactly invariance to row-specific baselines. It captures a feature shared by several causal procedures, but it supplies neither causal expected-utility maximisation nor dominance.

8. A PAIRWISE REDUCTION UNDER COMMON AFFINE INVARIANCE

The first cross-problem result shows that, with common affine invariance as the only invariance, the axioms of Definition 7.5 behave like the within-problem criteria: every inconsistency has a core of size one or two.

THEOREM 8.1 (Pairwise reduction). *On the domain of all finite real square matrices, under principal restriction, the inclusion-minimal inconsistent subsets of* {N, F, P, L, D, J, B, S, C, O, Q} *are exactly*

$$\{\mathsf{L}\}, \quad \{\mathsf{J}\}, \quad \{\mathsf{P}, \mathsf{D}\}, \quad \{\mathsf{P}, \mathsf{B}\}, \quad \{\mathsf{P}, \mathsf{S}\}, \quad \{\mathsf{P}, \mathsf{C}\},$$

and the maximal consistent subsets are exactly

maximal consistent subset	witness rule
{N, F, P, O, Q}	rank by column sums $\sum_k M_{ki}$
{N, F, D, B, S, C, O, Q}	rank by diagonal entries M_{ii}

The proof consists of five obstructions and a containment argument.

PROPOSITION 8.2 (Restricted Pareto alone). *No rule satisfies* L.

Proof. Consider

$$K = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

In rows 1 and 2, column 1 reads $(0, 1)$ and column 2 reads $(-1, 0)$, so L forces $1 \succ 2$. Likewise rows 2, 3 force $2 \succ 3$ and rows 1, 3 force $3 \succ 1$. Every option has a strict defeater, so $G(K) = \emptyset$. □

If bare cyclic relations without a nonempty recommendation were allowed when O is absent, the obstruction would instead be the pair $\{\mathsf{L}, \mathsf{O}\}$; the convention of Definition 7.4 makes it a singleton.

Proposition 8.3 (Global ratifiability alone). *No rule satisfies* J.

Proof. In $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ neither option maximises its own row. This is Death in Damascus in reduced form. □

Proposition 8.4 (Pareto against the diagonal). *No rule satisfies* P *and* D.

Proof. In $N_0 = \left(\begin{smallmatrix} 2 & 3 \\ 0 & 1 \end{smallmatrix}\right)$, column 2 strictly dominates column 1, so P gives $2 \succ 1$ and $G(N_0) = \{2\}$, while option 1 uniquely maximises the diagonal. This is Newcomb's problem in reduced form. □

Proposition 8.5 (Pareto against self-sovereignty; Briggs 2010). *No rule satisfies* P *and* S*, and no rule satisfies* P *and* B.

Proof. Let

$$M^{+} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}, \qquad M^{-} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}.$$

In M^{+}, column 1 weakly dominates column 2, strictly in row 3, so P gives $1 \succ 2$; in M^{-} it gives $2 \succ 1$. Rows 1 and 2 are the same in both matrices, contradicting S; so is the principal block on $\{1, 2\}$, contradicting B. □

This is an explicit instance of Briggs's theorem, which states the incompatibility of a Pareto condition and self-sovereignty for decision-theoretic rankings and does not require neutrality or transitivity (Briggs 2010, pp. 21–26). Causal decision theory satisfies her Pareto condition and violates self-sovereignty; evidential decision theory and Wedgwood's benchmark theory (Wedgwood 2013) satisfy self-sovereignty and violate Pareto. The conflict needs at least three options: on two-option problems the benchmark theory satisfies both. The pair $\{\mathsf{P}, \mathsf{S}\}$ is therefore not new.

Proposition 8.6 (Pareto against contraction). *No rule satisfies* P *and* C.

Proof. Return to the matrix K of Proposition 8.2. Its binary restrictions are

$$K|_{\{1,2\}} = K|_{\{2,3\}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad K|_{\{1,3\}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and P forces $G(K|_{\{1,2\}}) = \{1\}$, $G(K|_{\{2,3\}}) = \{2\}$, $G(K|_{\{1,3\}}) = \{3\}$. Each option of K is rejected in one binary restriction containing it, so by C none can be chosen from K. □

The mechanism is the classical binary cycle that makes contraction consistency incompatible with majority-type choice (Brandt and Harrenstein 2011; Sen 1970); the

modelling choice that gives it force here is that restriction deletes the rows of the removed options together with their columns. Section 10 shows that it disappears when the rows are retained.

Proof of Theorem 8.1. The six sets are inconsistent by Propositions 8.2–8.6, and each proper subset of each pair is consistent by the witnesses. Column sums satisfy N, F (scores are multiplied by a and shifted by $|I|b$), P (a weakly dominant column with one strict entry has a strictly larger sum), O, and Q. Diagonal ranking satisfies N, F, D, B (and hence S and C by Lemma 7.6), O, and Q. A consistent set contains neither L nor J. If it contains P, it excludes D, B, S, and C, and lies in the column-sum set; otherwise it lies in the diagonal set. □

Two of the reference rules one might expect to appear do not. Maximin over rows violates the strong Pareto axiom: in $\left(\begin{smallmatrix} 0 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, column 1 weakly dominates column 2, but both column minima are 0. Positive fixed-weight column averages satisfy P, and uniform weights add N, but fixed unequal weights require persistent labels or additional input. The selector T^{ν} of Proposition 5.5 is in general not a rule in the sense of A0: it uses structural information that M_D does not retain.

With common affine invariance alone, then, there is no Arrow-style family among these axioms.

9. Row-baseline invariance and contraction

Adding row-translation invariance changes the picture. It produces a genuine impossibility among four independent axioms, none of which concerns Pareto, ordering, or ex-ante value.

9.1. The circulant theorem

The proof embeds an arbitrary matrix in a matrix with a transitive symmetry group. It rests on a distinct-differences lemma of the kind used by Fink and Ruiz (1986) to embed graphs in circulants.

Lemma 9.1 (Distinct differences). *Let $k \geq 2$, $v_i = 2^i$ for $i = 0, \dots, k-1$, and $m > 2(2^{k-1} - 1)$. The $k(k-1)$ ordered differences $v_j - v_i$, $i \neq j$, are distinct and nonzero modulo m.*

Proof. For $j > i$, $v_j - v_i = 2^i(2^{j-i} - 1)$; the power of two and the odd part determine i and $j - i$, so the positive differences are distinct integers, and so are their negatives. Every difference has absolute value at most $D = 2^{k-1} - 1$, so two distinct differences differ by a nonzero integer of absolute value at most $2D < m$, and no difference is divisible by m. □

LEMMA 9.2 (Circulant embedding). *Every matrix $Z \in \mathbb{R}^{k\times k}$ with zero diagonal is a principal submatrix of a circulant matrix $A_{uv} = f(v-u \bmod m)$ on $\mathbb{Z}/m\mathbb{Z}$ with $f(0) = 0$.*

Proof. With v_i and m as in Lemma 9.1, set $f(0) = 0$ and $f(v_j - v_i) = Z_{ij}$ for $i \neq j$, which is well defined by the lemma, and give f arbitrary values on the unused residues. Then $A_{v_i v_j} = f(v_j - v_i) = Z_{ij}$. □

THEOREM 9.3 (Neutrality, row translations, and contraction force universal choice). *On the domain of all finite real square matrices, every rule satisfying* N, R, *and* C *(principal restriction) has $G(M) = I$ for every matrix (I, M). Consequently* {N, R, C, Q} *is inconsistent.*

Proof. Let (I, M) be given with $|I| = k \geq 2$ (the case $k = 1$ is trivial), and let $Z = Z(M)$, which has zero diagonal; by R, $G(Z) = G(M)$. By Lemma 9.2 and N (relabelling the vertices of the circulant so that the embedded ones carry the labels of I), Z is the principal restriction to I of a matrix A whose symmetry group acts transitively on its indices: each cyclic shift $u \mapsto u + 1$ of $\mathbb{Z}/m\mathbb{Z}$, transported by the relabelling, maps A to itself. By N, $G(A)$ is invariant under these symmetries, and since it is nonempty it contains every index. By C, $G(Z) = G(A|_I) \supseteq G(A) \cap I = I$. Hence $G(M) = I$, and no two matrices on the same index set have different choice sets. □

Without O the conclusion is universal *choice*, not universal indifference: incomparabilities may remain. The symmetry step is standard in the theory of tournament solutions, where neutral, nonempty solutions must select every vertex of a vertex-transitive tournament (Brandt 2011; Brandt et al. 2016); the embedding has the precedent noted above. What the theorem adds is the use of contraction to carry universal choice from symmetric matrices to all matrices, and of row-baseline invariance to reach every matrix from a zero-diagonal one. No published statement of this conclusion was found in the literature searched; its ingredients are established, and no priority is claimed.

PROPOSITION 9.4 (Independence). *Each three-element subset of* {N, R, C, Q} *is satisfiable, so the family is minimally inconsistent:*

omitted axiom	rule satisfying the other three
N	responsive fixed-label rule
R	diagonal ranking
C	row-range ranking, $r_i = \max_j M_{ij} - \min_j M_{ij}$
Q	universal indifference

All four rules also satisfy F *and* O.

Proof. Row ranges are unchanged by row translations, multiplied by a under F, and permuted under relabelling; the rule is nonconstant. Diagonal ranking satisfies C by Lemma 7.6. The *responsive fixed-label rule* fixes two labels u, v and a well-order of $\mathbb{L}$ in which u, v form an adjacent block above all other labels. When both occur, u and

v are compared by the sign of $M_{uu} - M_{uv}$ (a tie if it is zero); all other comparisons follow the fixed order. The comparison of u and v uses a within-row difference of their principal block, so the rule satisfies R, F, and B, hence C; it is responsive, so Q holds, but it is not neutral. □

If neutrality is regarded as part of the reduction A0 rather than as an axiom, the same result is a minimal three-axiom family {R, C, Q} relative to that background.

PROPOSITION 9.5. {N, R, B, Q} *is minimally inconsistent, and so is* {R, D}.

Proof. Suppose $i \succ_M j$ in some matrix. By B the same holds in $M|_{\{i,j\}}$, and by R in its row normalisation $H = \left(\begin{smallmatrix} 0 & x \\ y & 0 \end{smallmatrix}\right)$, with i first. Let

$$C = \begin{pmatrix} 0 & x & y \\ y & 0 & x \\ x & y & 0 \end{pmatrix}.$$

Its principal blocks on $(1, 2)$, $(2, 3)$, and $(3, 1)$ all equal H, so N and B give $1 \succ 2$, $2 \succ 3$, and $3 \succ 1$, and $G(C) = \emptyset$. Hence no strict comparison ever occurs and $G(M) = I$ always, contradicting Q. Minimality follows from the witnesses of Proposition 9.4, with row-range ranking now serving as the rule that omits B. For {R, D}: adding 2 to the second row of $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ changes the unique diagonal maximiser from option 1 to option 2, while R keeps the choice fixed. □

Unlike Theorem 9.3, this argument uses the link between comparisons and nonempty maximal choice fixed in Definition 7.4.

THEOREM 9.6 (Map with row translations). *Adding* R *to the family of Theorem 8.1 introduces exactly three new minimal inconsistent subsets,*

$$\{\mathsf{R}, \mathsf{D}\}, \qquad \{\mathsf{N}, \mathsf{R}, \mathsf{B}, \mathsf{Q}\}, \qquad \{\mathsf{N}, \mathsf{R}, \mathsf{C}, \mathsf{Q}\},$$

and the maximal consistent subsets become

witness rule	maximal consistent subset
column sums	{N, F, P, R, O, Q}
diagonal ranking	{N, F, D, B, S, C, O, Q}
row ranges	{N, F, R, S, O, Q}
responsive fixed-label rule	{F, R, B, S, C, O, Q}
universal indifference	{N, F, R, B, S, C, O}

Proof. Column sums satisfy R because a row translation shifts every column sum by $\sum_k b_k$; row ranges satisfy S because each score depends only on its own row. Let a set avoid all minimal inconsistent subsets. If it contains P, its pairwise conflicts confine it to the first row. If it omits P and R, it fits the second. If it contains R, it omits D; if it contains both N and Q, the two four-element obstructions exclude B and C, leaving the third row; without N it fits the fourth, and without Q the fifth. □

9.2. An exact domain condition

Theorem 9.3 needs matrices of unbounded size: with a fixed maximum size n, the rule that chooses everything below size n and maximises row ranges at size n satisfies N, R, C, and Q. The exact requirement on the domain can be stated without quantifying over rules. Let $\mathcal{D}$ be a set of matrices closed under relabelling, and interpret each invariance as applying whenever both the matrix and its transform belong to $\mathcal{D}$. Form a directed graph on the *pointed matrices* $\mathcal{P}_{\mathcal{D}} = \{(M, i) : M \in \mathcal{D},\ i \in I_M\}$, with

(i) edges in both directions between (M, i) and $(\sigma \cdot M, \sigma(i))$ for admissible relabellings;

(ii) edges in both directions between (M, i) and $(M + b\mathbf{1}^\top, i)$ for admissible row translations;

(iii) extension edges $(M|_K, i) \to (M, i)$ for $i \in K$, whenever $M|_K$ and M are in $\mathcal{D}$.

Let $\mathrm{Reach}(x)$ be the set of pointed matrices reachable from x (including x), and let $\Phi_M = \{(M, i) : i \in I_M\}$ be the fibre of M.

Theorem 9.7 (Exact domain criterion). *On $\mathcal{D}$, the axioms* N, R, C, Q *are jointly inconsistent if and only if*

$$\text{for every } x \in \mathcal{P}_{\mathcal{D}} \text{ there is } M \in \mathcal{D} \text{ with } \Phi_M \subseteq \mathrm{Reach}(x). \tag{DR}$$

Proof. Say (M, i) is *rejected* if $i \notin G(M)$. Rejection propagates along relabelling and translation edges by N and R, and along extension edges by the contrapositive of C. Under (DR), a single rejection at x would reject the entire fibre of some M, contradicting $G(M) \neq \emptyset$; so nothing is ever rejected, and Q fails. Conversely, if (DR) fails at $x = (M_0, i_0)$, set

$$G_x(M) = \{i : (M, i) \notin \mathrm{Reach}(x)\}.$$

Each $G_x(M)$ is nonempty because no fibre lies in $\mathrm{Reach}(x)$. Since relabelling and translation edges are bidirectional, $\mathrm{Reach}(x)$ is closed under them, which gives N and R. If $i \in G_x(M) \cap K$ but $i \notin G_x(M|_K)$, the extension edge would put (M, i) in $\mathrm{Reach}(x)$; so C holds. Finally $i_0 \notin G_x(M_0)$, so $G_x(M_0)$ is a nonempty proper subset of I_{M_0}, and some relabelling of M_0 on the same index set moves it, which gives Q. Tying chosen options above rejected ones yields O as well. □

The circulant condition is sufficient for (DR) but not necessary. If F is also imposed, admissible affine transformations are added as bidirectional edges.

9.3. Physical restrictions on problems

Does the richness that (DR) demands survive natural restrictions on the problems themselves?

Proposition 9.8 (Bounded payoffs). *On the domain of $[0, 1]$-valued matrices, $\{$*N, R, C, Q*$\}$ is consistent; adding* F *makes it inconsistent again.*

Proof. Fix $r \in [\frac{1}{2}, 1)$ and put $i \to j$ if $M_{ii} - M_{ij} > r$ and $M_{ji} - M_{jj} > r$. An edge forces $M_{ii} > r \geq \frac{1}{2} \geq 1 - r > M_{jj}$, so no vertex has both an incoming and an outgoing edge and the graph is acyclic. Choose the options with no incoming edge, tied above the rejected ones, which are tied among themselves. The conditions use only within-row differences, giving R; edges depend only on principal 2×2 blocks and persist under restriction, giving C; the rule is neutral, ordered, and responsive (it chooses $\{1\}$ from $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ and both options from the zero matrix); and raising a diagonal entry only removes incoming and adds outgoing edges, so it satisfies Mon. It fails F, since scaling by a small $a > 0$ removes every edge.

With F, given $M \in [0,1]^{n \times n}$ and small $\alpha > 0$, pass from M to αM by F and then to $A_{ij} = \frac{1}{2} + \alpha(M_{ij} - M_{ii})$ by R; every intermediate matrix stays in $[0,1]$. Embed A in a circulant as in Lemma 9.2, filling unused residues with $\frac{1}{2}$, and run the proof of Theorem 9.3. □

Proposition 9.9 (Prediction bounded away from accuracy). *Let $P_n = (1-\eta)I + \frac{\eta}{n}\mathbf{1}\mathbf{1}^\top$ with $0 < \eta < 1$, a prediction channel whose probability of a correct forecast is $1-\eta+\eta/n \leq 1-\eta/2$ for $n \geq 2$. With unrestricted real payoffs, every $M \in \mathbb{R}^{n \times n}$ is the causal matrix of a problem whose predictor uses P_n. With payoffs in $[0,1]$ and F, the circulant argument survives, uniformly in n.*

Proof. P_n is invertible, with $P_n^{-1} = \frac{1}{1-\eta} I - \frac{\eta}{(1-\eta)n}\mathbf{1}\mathbf{1}^\top$. Let the state be the forecast label, drawn from row i of P_n when i is chosen, and let $U = P_n^{-1} M$ be the state-by-action payoff table; then the causal matrix is $P_n U = M$. Since $P_n^{-1}\mathbf{1}\mathbf{1}^\top = \mathbf{1}\mathbf{1}^\top$, the constant matrix $\frac{1}{2}\mathbf{1}\mathbf{1}^\top$ is mapped to itself, and the absolute row sums of P_n^{-1} are at most $(1+\eta)/(1-\eta)$; so a neighbourhood of the constant matrix whose radius does not depend on n is mapped into $[0,1]$, and the compressed matrices of Proposition 9.8 remain realisable. □

An accuracy bound alone, then, does not destroy richness. Completely uninformative prediction is different: if all induced laws coincide, the channel loses rank, every row of the causal matrix is the same, and the problem is an ordinary causal one.

Proposition 9.10 (Bounded action sets). *For every $k \geq 2$, the symmetric extensions required by Theorem 9.3 are realisable by problems with perfect recall and at most k actions per information set, provided contraction applies to arbitrary sets of policies.*

Proof. Take a complete binary decision tree of depth m, with the agent observing its own past actions and the prediction hidden until the end. It has $2^m - 1$ information sets and $N = 2^m$ action paths, and each path is produced by exactly $L = 2^{2^m - 1 - m}$ policies, since prescriptions off the path are free. Let a predictor forecast the eventual path and let the payoff depend on the forecast and the actual path; any path matrix C is then realisable, and the policy matrix is $C \otimes J_L$, with J_L the all-ones $L \times L$ matrix. If C is circulant, rotating path blocks and permuting duplicate policies within blocks act transitively on the policies. Any target matrix embeds in a circulant whose size is

a power of two (Lemma 9.2), and is recovered by restricting to one representative policy from each designated block. □

The proviso matters: arbitrary sets of surviving policies are generally not obtained by pruning local action sets, which produces product-shaped sets. Restricting contraction to literal pruning is a different axiom (Section 10).

PROPOSITION 9.11 (A single bounded choice). *With one information set and an unbounded action set, every square matrix is realisable. With one information set and at most $k \geq 2$ actions, the rule that is universally indifferent below size k and ranks by $M_{ii} - \frac{1}{k}\sum_j M_{ij}$ at size k satisfies* N, F, R, C, O, Q, *and* Mon.

Proof. The first claim is Theorem 7.2. For the second, every proper restriction of a size-k matrix lands where every option is chosen, so C holds; the score is a within-row difference, invariant under F up to scale, neutral, responsive, and increasing in M_{ii}. □

No finite size ceiling, therefore, forces triviality under the existential form of Q. If non-triviality is required specifically on binary menus, the three-option circulant $\left(\begin{smallmatrix} 0 & x & y \\ y & 0 & x \\ x & y & 0 \end{smallmatrix}\right)$ already blocks it as soon as three options are available.

10. The semantics of a smaller menu

Theorems 8.1 and 9.3 suggest a dichotomy: menu-coherent evaluation forces ex-ante evaluation, while causal evaluation cannot be menu-coherent without becoming trivial. Both theorems, however, rest on one way of removing options from a problem: principal restriction, which deletes the removed options' rows together with their columns. This section examines that step. It turns out to be where the dichotomy fails.

10.1. Three operations

A smaller menu can be modelled in at least three ways.

Principal restriction (C_{pr}). Delete the removed options' rows and columns. A smaller menu then also removes the beliefs that the agent would have if it chose the removed options.

Column restriction ($\mathsf{C}_{\mathrm{col}}$). Keep every row and delete only the columns. The input becomes a rectangular matrix of belief states by available options.

Fixed-reference restriction ($\mathsf{C}_{\mathrm{ref}}$). Evaluate options by averaging rows with a reference distribution over belief states, as in fixed-prior causal decision theory, and keep that distribution when options are removed.

PROPOSITION 10.1 (Law-preserving removal is principal). *Let D' be obtained from D by making only the policies in $K \subseteq \Pi_D$ available, and suppose that their response*

laws and consequences are unchanged: $\mu_i' = \mu_i$ *and* $U'(s,i) = U(s,i)$ *for* $i \in K$. *Then* $M_{D'} = M_D|_{K\times K}$.

Proof. Immediate from Definition 7.1: each entry of $M_{D'}$ involves only laws and consequences of policies in K. □

Principal restriction is thus forced by the definition of the causal matrix once law-preserving removal is stipulated. Two qualifications apply. First, arbitrary subsets K do not arise by pruning local action sets, which produces product-shaped sets of policies; an arbitrary K is realised by treating the surviving policies as macro-actions of a new one-choice predictive problem, which preserves the response laws but not the extensive form. Second, an intervention that actually announces a smaller menu may change how a predictor responds; the model does not require $\mu_i^{D'} = \mu_i^D$, and law preservation is an additional cross-problem assumption. Removal by macro-actions must also be distinguished from commitment before prediction (Definition 5.1), which changes the counterfactual structure and need not yield a principal submatrix. Spencer and Wells (2019) make the related point that converting a simultaneous problem into a sequential one can create a different decision problem, and Gallow (2026) explicitly defines a binary sub-decision by removing the other options *and conditioning credences on the surviving disjunction*, which is yet another operation.

10.2. Rectangular and fixed-reference domains

Definition 10.2 (Rectangular inputs). A *rectangular input* is a triple (J, A, M) with $J \subseteq \mathbb{L}$ finite, $\emptyset \neq A \subseteq J$, and $M \in \mathbb{R}^{J\times A}$. Here J indexes the retained belief perspectives and A the available options, and option $a \in A$ has its *own row* a. Column restriction is $(J, A, M)|_K = (J, K, M_{J\times K})$ for $\emptyset \neq K \subseteq A$.

The own-row association is what keeps diagonal optimality, self-sovereignty, and monotonicity meaningful. The off-menu rows are retained descriptions of the old conditional or counterfactual laws; they are additional data, not rows of the canonical causal matrix of the smaller problem, which remains square. Column restriction can therefore describe a fair augmented problem, but it does not follow from reduction to $M_{D'}$ alone.

The axioms extend to rectangular inputs as follows: relabellings act on J (and hence on A); R adds b_j to row j for every $j \in J$; P quantifies over all rows in J; D maximises M_{aa} over $a \in A$; S compares inputs with the same J and A; Q asks for two inputs with the same J and A and different choice sets; and contraction becomes

$$\mathrm{C}_{\mathrm{col}} : \qquad G(J, A, M) \cap K \subseteq G(J, K, M_{J\times K}).$$

Binary independence splits into two inequivalent axioms:

- B_{pr} the comparison of a, b agrees with their comparison in the principal square input $(\{a, b\}, \{a, b\}, M_{\{a,b\}\times\{a,b\}})$;
- $\mathrm{B}_{\mathrm{col}}$ the comparison of a, b agrees with their comparison in $(J, \{a, b\}, M_{J\times\{a,b\}})$, which keeps every row.

Both imply C_{col}, by the argument of Lemma 7.6; B_{pr} implies S, while B_{col} does not, because it never compares inputs with different belief universes.

For fixed-reference restriction, supply a probability vector $w \in \Delta(J)$, evaluate $v_a(M, w) = \sum_j w_j M_{ja}$, and restrict by $(J, A, M, w)|_K = (J, K, M_{J\times K}, w)$; under relabelling, w is relabelled too, and uniform weights give a rule that depends on the rectangular matrix alone. A reference distribution over belief perspectives need not be the agent's probability over its *available* choices: if it is meant to be, eliminating choices ordinarily requires conditioning, and keeping positive weight on eliminated choices requires a different interpretation. The next result shows that the reference cannot be updated while preserving contraction.

THEOREM 10.3 (Weight preservation). *Let w, w' be probability vectors on the same set J, let at least two columns survive a restriction, and suppose the update $w \mapsto w'$ does not depend on the payoff entries. Weighted averaging ranks every payoff matrix identically under w and w' if and only if $w' = w$, and contraction holds for every payoff matrix under the update only if $w' = w$.*

Proof. The difference of two columns can be any $x \in \mathbb{R}^J$, so identical rankings require $\operatorname{sign}(w^\top x) = \operatorname{sign}(w'^\top x)$ for all x; the two functionals are then positive multiples of each other, and normalisation gives $w' = w$. If $w' \neq w$, choose x with $w^\top x = 0 < w'^\top x$ and replace it by $x - \epsilon\mathbf{1}$ for small $\epsilon > 0$, so that $w^\top x < 0 < w'^\top x$. Realise x as the difference of two surviving columns a, b, so that b is preferred under w and a under w', and set every removed column low enough that b is a full-menu maximiser under w. After restriction b is no longer chosen, and contraction fails. □

Deleting rows and renormalising their weights therefore preserves all rankings only when the deleted rows had total weight zero; with full-support weights, any proper row deletion changes the reference.

operation	available columns	belief rows and reference	canonical law-preserving matrix?
principal	delete unavailable options	delete the corresponding rows	yes, given law preservation
column	delete unavailable options	retain every row	needs augmented reference data
fixed reference	delete unavailable options	retain J and w	needs a supplied reference
delete and renormalise	delete unavailable options	change the reference	a different belief update

10.3. Consistency maps under column restriction

THEOREM 10.4 (Column map with principal binary independence). *On all finite rectangular inputs, the inclusion-minimal inconsistent subsets of*

$$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{P}, \mathsf{D}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$$

are exactly

$$\{\mathsf{P}, \mathsf{D}\}, \quad \{\mathsf{P}, \mathsf{B}_{\mathrm{pr}}\}, \quad \{\mathsf{P}, \mathsf{S}\}, \quad \{\mathsf{R}, \mathsf{D}\}, \quad \{\mathsf{N}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{Q}\},$$

and the maximal consistent subsets are

maximal consistent subset	witness
$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{P}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	column sums $\sum_{j \in J} M_{ja}$
$\{\mathsf{N}, \mathsf{F}, \mathsf{D}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	diagonal ranking
$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	full-menu relative scores
$\{\mathsf{F}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	responsive fixed-label rule
$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}\}$	universal indifference

All the witnesses also satisfy Mon.

Proof. The *full-menu relative-score rule* ranks by $s_a = M_{aa} - \frac{1}{|A|} \sum_{c \in A} M_{ac}$ when $A = J$ and ties every option when $A \subsetneq J$. It is neutral and invariant under F and R; on full menus each score depends only on its own row, and inputs compared by S have the same J and A, so S holds; every proper column restriction produces a non-full menu, on which everything is chosen, so $\mathsf{C}_{\mathrm{col}}$ holds; and s_a increases with M_{aa}. Column sums are unchanged on surviving columns by column restriction, which gives $\mathsf{C}_{\mathrm{col}}$, and shift uniformly under R. The other witnesses are as in Theorem 9.6.

The obstructions $\{\mathsf{P}, \mathsf{D}\}$, $\{\mathsf{P}, \mathsf{S}\}$, and $\{\mathsf{R}, \mathsf{D}\}$ are proved as in Propositions 8.4, 8.5, and 9.5, with square inputs; $\{\mathsf{P}, \mathsf{B}_{\mathrm{pr}}\}$ follows from $\mathsf{B}_{\mathrm{pr}} \Rightarrow \mathsf{S}$; and $\{\mathsf{N}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{Q}\}$ is proved by the three-cycle of Proposition 9.5, applied to square inputs. For completeness: a set containing P excludes D, B_{pr}, and S, and the column-sum witness covers it. Without P and without R, diagonal ranking covers it. With R, D is excluded and at least one of N, B_{pr}, Q is omitted, and the last three witnesses cover these cases. □

The obstruction $\{\mathsf{P}, \mathsf{C}\}$ of Theorem 8.1 has disappeared, and so has $\{\mathsf{N}, \mathsf{R}, \mathsf{C}, \mathsf{Q}\}$. The matrix

$$W = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

makes the difference visible. Its column sums tie. Under principal restriction, P successively forces 1 over 2, 2 over 3, and 3 over 1 in the binary sub-menus. Under column restriction, the third row remains in every binary sub-menu and blocks those Pareto judgements, and column averaging continues to tie.

THEOREM 10.5 (Column map with column binary independence). *Replacing* B_{pr} *by* $\mathsf{B}_{\mathrm{col}}$*, the minimal inconsistent subsets are exactly*

$$\{\mathsf{P}, \mathsf{D}\}, \qquad \{\mathsf{P}, \mathsf{S}\}, \qquad \{\mathsf{R}, \mathsf{D}\},$$

and the maximal consistent subsets are $\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{P}, \mathsf{B}_{\mathrm{col}}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$ *(column sums),* $\{\mathsf{N}, \mathsf{F}, \mathsf{D}, \mathsf{B}_{\mathrm{col}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$ *(diagonal ranking), and* $\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{B}_{\mathrm{col}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$ *(the binary-universe rule below).*

Proof. The *binary-universe rule* ranks by $s_a = M_{aa} - M_{ab}$ when $J = A = \{a, b\}$ and ties everything when $|J| \geq 3$; singleton menus are trivial. It satisfies Q on a two-row universe, and $\mathsf{B}_{\mathrm{col}}$ never connects different belief universes, so it does not force the two-row behaviour to extend to three rows. Column sums satisfy $\mathsf{B}_{\mathrm{col}}$ because deleting other columns leaves the two sums unchanged. The containment argument is as before: P excludes D and S; without P and R, diagonal ranking applies; with R and without P, the binary-universe rule applies. □

The weakness here lies in the existential form of Q. Fix a belief universe J with $|J| \geq 3$, allow all its menus, and let Q_J require non-triviality on that universe.

THEOREM 10.6 (A five-axiom family). $\{\mathsf{N}, \mathsf{R}, \mathsf{B}_{\mathrm{col}}, \mathsf{S}, \mathsf{Q}_J\}$ *is a minimal inconsistent family.*

Proof. If every binary comparison on J were a tie, $\mathsf{B}_{\mathrm{col}}$ would make every comparison on J a tie, and every menu would be chosen entire, contradicting Q_J; so some strict binary comparison occurs on J. On a binary menu $\{a, b\}$, S makes the comparison depend only on the principal block of a and b, and R lets its diagonal be normalised to zero. Neutrality, applied to the relabellings of J that permute three of its elements cyclically, replicates the strict comparison around a three-cycle of binary menus, and $\mathsf{B}_{\mathrm{col}}$ transfers the three comparisons to the menu of those three options, all of which are then strictly beaten. The witnesses omitting N, R, $\mathsf{B}_{\mathrm{col}}$, S, Q_J are, respectively, the responsive fixed-label rule, diagonal ranking, full-menu relative scores, column sums, and universal indifference; they can additionally satisfy F, $\mathsf{C}_{\mathrm{col}}$, O, and Mon. □

The strongest replacement thus depends on whether non-triviality is required on a universe large enough to support the obstruction.

10.4. What survives of the circulant theorem

PROPOSITION 10.7. *Uniform column averaging satisfies* N, F, R, P, $\mathsf{B}_{\mathrm{col}}$, $\mathsf{C}_{\mathrm{col}}$, O, Q, *and* Mon. *The same holds for weighted averaging with any fixed full-support reference* w *under* $\mathsf{C}_{\mathrm{ref}}$ *(with* N *understood to relabel* w*).*

Proof. Each option's score is unaffected by the other columns and by their removal; row translations shift all scores equally; positive weights give P; raising M_{aa} raises only the score of a. □

Under column or fixed-reference restriction, then, no impossibility can be obtained from any subset of these axioms beyond those listed in Theorems 10.4 and 10.5. In particular, row-baseline-invariant evaluation *can* be menu coherent when the belief model is held fixed, and the first half of the proposed dichotomy fails. The impossibility of Theorem 9.3 concerns the preservation of comparisons when the

belief perspectives associated with removed options disappear, not the preservation of choices when options disappear.

11. What characterises ex-ante evaluation

The second half of the dichotomy claimed that menu coherence forces ex-ante evaluation. This section shows that principal binary independence does force it, given symmetry, ordering, and monotonicity, and that contraction does not.

11.1. Binary independence: the Self-Esteem theorem

Theorem 11.1 (Relational Self-Esteem theorem). *On the domain of all finite real square matrices, a rule satisfies* N, B_{pr}, *and* O *if and only if there is a complete transitive relation* $\succeq^*$ *on* $\mathbb{R}$ *such that, for every* M *and all* i, j,

$$i \succeq_M j \iff M_{ii} \succeq^* M_{jj}.$$

Proof. Consider

$$H = \begin{pmatrix} p & r & p \\ q & s & u \\ p & t & p \end{pmatrix}.$$

The principal block on $\{1,3\}$ is constant, so the transposition of 1 and 3 fixes it; by N and B_{pr} the comparison of 1 and 3 is symmetric, and by completeness $1 \sim 3$. By transitivity, 1 and 3 compare in the same way with 2. By B_{pr} the first comparison is that of the first index with the second in $\left(\begin{smallmatrix} p & r \\ q & s \end{smallmatrix}\right)$, and the second is that of the first index with the second in the principal block on $(3,2)$, which is $\left(\begin{smallmatrix} p & t \\ u & s \end{smallmatrix}\right)$. Since r, q, t, u are arbitrary, the comparison of two options does not depend on the off-diagonal entries of their block, and by B_{pr} and N it is a function of the ordered pair of diagonal values. Define $x \succeq^* y$ if the first index is weakly preferred in $\mathrm{diag}(x, y)$. Completeness is inherited, and transitivity follows by applying B_{pr} to $\mathrm{diag}(x, y, z)$. Conversely, ranking by a weak order of the diagonal values satisfies N, B_{pr}, and O, and has nonempty maximal sets on finite menus. □

This is the relational form of the "Self-Esteem Theorem" of Podgorski (2022): if the comparison of two options is a fixed function of the four values $V_X(X)$, $V_Y(Y)$, $V_X(Y)$, $V_Y(X)$ and overall verdicts respect independence of irrelevant alternatives, the comparison depends only on each option's value by its own lights, $V_X(X)$ and $V_Y(Y)$. Podgorski concludes that among overall-value views only the evidential one respects that independence, and his tournament decision theory escapes the theorem by producing no single overall value. Here the common comparison function is supplied by N and B_{pr}, and the scalar overall value by O; the relational version assumes no real-valued representation of $\succeq^*$.

Corollary 11.2. *Adding* F *leaves exactly three rules: increasing diagonal ranking, decreasing diagonal ranking, and universal indifference. Adding* Q *and* Mon *leaves only increasing diagonal ranking.*

Proof. Every pair $x > y$ is mapped to every other such pair by a positive affine transformation, so under F all unequal pairs are oriented alike; equal values tie by reflexivity and N. Q excludes indifference. Under decreasing ranking, raising M_{11} in the zero 2×2 matrix removes option 1 from the choice set, violating the choice clause of Mon. □

Since $M_{ii} = V_D(i)$,

$$\mathsf{N} + \mathsf{F} + \mathsf{B}_{\mathrm{pr}} + \mathsf{O} + \mathsf{Q} + \mathsf{Mon}$$

characterises ex-ante policy ranking under the stated matrix semantics. Under the structural axioms N, F, B_{pr}, O, a single binary benchmark with unequal diagonal entries identifies which of the three rules applies; no finite benchmark, however, certifies that the structural axioms themselves hold everywhere.

Proposition 11.3 (Exact domain threshold). *For domains consisting of all matrices up to a maximum size, the characterisation holds if and only if matrices of size three are included.*

Proof. With every matrix of sizes one to three, the proof of Theorem 11.1 goes through, and B_{pr} propagates the conclusion to any larger sizes present. With sizes at most two, B_{pr} is vacuous, and ranking by $s_i = M_{ii} - M_{ij}$ satisfies N, F, O, Q, and Mon; on $\left(\begin{smallmatrix} 1 & 100 \\ 0 & 0 \end{smallmatrix}\right)$ it chooses option 2. □

Merely having some three-option problems is not enough; a sufficient richness condition is that the domain contain all binary matrices, all the clone extensions H above, and their relabellings (and, for the affine conclusion, the relevant affine transforms). The argument also works on $[0, 1]$-valued matrices of sizes up to three, with F interpreted internally.

11.2. Contraction does not force the diagonal

Can contraction replace binary independence? Under every menu semantics, the answer is no.

Theorem 11.4 (A contraction-consistent family without diagonal optimality). *Write $d_i = M_{ii}$. For $\lambda \geq 0$ let $i \to_\lambda j$ if $d_i - d_j > \lambda |M_{ij} - M_{ji}|$; choose the options with no incoming edge, tied at the top, and rank the rejected options below them by their diagonal entries. Every such rule satisfies* N, F, C_{pr}, $\mathsf{C}_{\mathrm{col}}$, O, Q, *and* Mon. *At $\lambda = 0$ it is increasing diagonal ranking; for every $\lambda > 0$ it sometimes recommends an option whose diagonal entry is not maximal.*

Proof. Every edge strictly lowers the diagonal, so the edge relation is acyclic and the choice set is nonempty. Whether $i \to_\lambda j$ depends only on $M_{ii}, M_{jj}, M_{ij}, M_{ji}$, which survive both principal and column restriction; so a chosen option acquires no incoming edge under restriction, which gives both forms of contraction. Raising d_i can only remove edges into i and add edges out of i; hence i's chosen status can only improve, other options' chosen status can only worsen, and if i stays rejected its rank among

the rejected improves, which gives Mon. N and F are clear, since $|M_{ij} - M_{ji}|$ scales with a and is translation invariant. For $\lambda > 0$, in $\left(\begin{smallmatrix} 1 & 0 \\ 100/\lambda & 0 \end{smallmatrix}\right)$ there is no edge, and both options are chosen. □

Even an ordered, neutral, affine-invariant, monotone, contraction-consistent rule, therefore, need not maximise ex-ante value. The family also defeats finite certification: given finitely many benchmark matrices, choose $\lambda > 0$ so small that every option with non-maximal diagonal in every benchmark receives an edge from a diagonal maximiser. The whole ranking then coincides with diagonal ranking on every benchmark, while a new binary matrix with a large off-diagonal asymmetry relative to its diagonal gap violates D.

PROPOSITION 11.5 (Maximal cardinality). *Let* $\mathfrak{c} = |\mathbb{R}|$. *There are* $2^{\mathfrak{c}}$ *rules satisfying* N, F, C_{pr}, C_{col}, O, Q, *and* Mon.

Proof. For $E \subseteq \mathbb{R}$, put $i \to j$ if $d_i > d_j$, $s = M_{ji} - M_{ij} > 0$, and either $d_i - d_j > s$, or $d_i - d_j = s$ and $(d_j - M_{ij})/s \in E$; complete as in Theorem 11.4. Edges descend the diagonal and depend only on principal blocks, so the choice set is nonempty and both contractions hold. Raising d_i only decreases the gaps $d_k - d_i$, so it never creates an edge into i (at the boundary it may delete one), and only increases the gaps $d_i - d_j$, so it never deletes an edge out of i; this gives Mon. The boundary parameter $(d_j - M_{ij})/s$ is invariant under positive affine maps. On

$$M_r = \begin{pmatrix} r+1 & 0 \\ 1 & r \end{pmatrix}$$

we have $d_1 - d_2 = s = 1$ and boundary parameter r, so the choice is $\{1\}$ if $r \in E$ and $\{1, 2\}$ otherwise. Distinct sets E give distinct rules, and $2^{\mathfrak{c}}$ is the number of all rules on finite real matrices over a countable label universe. □

The class cannot, therefore, be parameterised by finitely many real numbers, or even countably many.

11.3. Fixed-reference scores

Suppose that only the score vector $v_a = \sum_j w_j M_{ja}$ is available, and that restriction deletes coordinates. Even then rigidity does not return. The rule choosing the top $k \geq 2$ scores, including boundary ties, $G_k(v) = \{a : \#\{b : v_b > v_a\} < k\}$, with the chosen options tied above the others and the others ranked by score, satisfies N, F, contraction, O, Q, and Mon.

PROPOSITION 11.6. *There are* $2^{\mathfrak{c}}$ *rules on finite score vectors satisfying* N, F, *contraction,* O, Q, *and* Mon, *already distinguished on vectors of length four.*

Proof. Fix a set E of normalised shapes of triples. For a candidate j and a triple T of other options, let $g = (v_i - v_j)_{i \in T}$. Reject j if some triple has $\sum g > 0$, or has $\sum g = 0$, $g \neq 0$, and sorted normalised shape in E; choose the rest, and complete the ranking

as above. A score maximiser is never rejected, since all its gaps are non-positive. Deletion removes witnesses, so contraction holds. Raising v_j lowers every gap sum for j, so a zero-sum rejection can only turn into a negative sum, and raises the relevant sums for other candidates; this gives Mon. Positive affine maps preserve signs and normalised shapes. On $v = (0, -1, t, 1-t)$ with $0 < t < \frac{1}{2}$, the option of score 0 has the single triple $(-1, t, 1-t)$, of sum zero, and is rejected exactly when its shape lies in E; these shapes are distinct for distinct t. □

With a fixed full-support w, raising M_{jj} raises only v_j, so the matrix form of Mon follows. The upshot is that the dichotomy fails in both directions: contraction under any of the three semantics is compatible with a rich family of non-ex-ante rules, and row-baseline-invariant evaluation is compatible with contraction once the belief model is fixed. What does force ex-ante evaluation is principal binary independence, which is a strong requirement: it imposes on every comparison the beliefs that the two compared options would induce, and no others.

12. Types, parameters, and certification

A recurring hope in the debate is that the relevant problem types are few: that a decision theory could be settled by its behaviour on a small catalogue of representative problems. Whether this is so depends on what counts as the same type, and on which structural axioms a theory is already known to satisfy.

12.1. Problems

Boolean classifications are automatically finite. For k local criteria, the profile $\mathcal{S}(D) = \{I \subseteq \{1, \dots, k\} : \text{some policy satisfies every criterion in } I\}$ takes at most 2^{2^k} values. A finite classification by joint-satisfiability profiles thus exists trivially; it need not help to solve any problem, and C4, being a relation between problems, is not one of these local predicates.

Exact recommendation structures are not finite. Call two problems equivalent if a bijection of their policy spaces preserves the set of policies acceptable under each criterion.

Proposition 12.1. *Even with two actions and one always-reached information set, the continuous mixed model contains infinitely many pairwise inequivalent problems.*

Proof. For distinct $r_1, \dots, r_n \in (0, 1)$, let a reward $Y \in \{0, 1\}$, independent of the realised action, satisfy $\Pr(Y = 1 \mid p) = 1 - \prod_i (p - r_i)^2$, a continuous probability since each factor lies in $[0, 1]$. The value is maximised exactly at $p \in \{r_1, \dots, r_n\}$, so exactly n policies satisfy C1, while every policy satisfies C2 and C3. A bijection preserving the C1-sets cannot identify problems with different n. □

For deterministic policies, Theorem 7.2 already shows that every finite recommendation structure occurs. The permissive definition of a theory has a further, direct consequence.

PROPOSITION 12.2 (No finite benchmark suite). *No finite set of problems can establish that an arbitrary decision theory satisfies* C1 *everywhere.*

Proof. Modify T^{ex} only on one two-action problem outside the suite, with a strictly better action, so as to select the worse action. The two theories agree on every benchmark, and only one satisfies C1 everywhere. □

This holds even if the benchmarks and the omitted problem are of the same intuitive type. Representative testing requires an invariance requirement on theories. Note also that infinitely many equivalence classes do not by themselves preclude a finite-dimensional parameterisation: a single real parameter indexes infinitely many cases.

12.2. Rules

The question is sharper for rules on matrices, where invariance axioms are available. Finitely many parameters must then be read cardinally: a finite-dimensional input space does not make an arbitrary rule on it a finite-dimensional family.

PROPOSITION 12.3. *There are* $2^{\mathfrak{c}}$ *rules satisfying* A0, N, F, *and* O. *Finite benchmark suites cannot certify* P *or* D *for them.*

Proof. For $0 < t < 1$ let $N_t = \left(\begin{smallmatrix} 0 & -1 \\ t+1 & t \end{smallmatrix}\right)$. These lie in distinct orbits of relabelling combined with positive affine maps, and each has trivial stabiliser: without a swap, preserving the entries 0 and -1 forces the identity map; with a swap, the diagonal $(t, 0)$ must be mapped to $(0, s)$, which requires a negative scale. For each $E \subseteq (0, 1)$, choose option 1 on N_t when $t \in E$ and option 2 otherwise, transport this ranking equivariantly over each orbit, and use diagonal ranking elsewhere. This gives $2^{\mathfrak{c}}$ distinct rules, whereas finite tuples of reals form a set of cardinality $\mathfrak{c}$. In N_t, column 1 strictly dominates column 2 and the second diagonal entry is larger; so modifying column sums, or diagonal ranking, on one untested orbit produces a violation of P, or of D, that no finite suite (even an adaptive one) detects. □

The construction uses the permission to have arbitrary lookup rules; it does not carry over unchanged to classes with continuity or measurability requirements.

PROPOSITION 12.4. *There are* $2^{\mathfrak{c}}$ *rules satisfying* N, F, *and* B *(without* O*), and finite suites cannot certify* D *for them.*

Proof. For $E \subseteq \mathbb{R}$ and $M_{ii} > M_{jj}$, declare $i \succ_M j$ if $(M_{ij} - M_{jj})/(M_{ii} - M_{jj}) \in E$; never prefer a lower diagonal strictly to a higher one, and tie all other pairs. The ratio is invariant under positive affine maps and depends only on the principal block, and strict comparisons descend the diagonal, so choice sets are nonempty. On $M_t = \left(\begin{smallmatrix} 1 & t \\ 0 & 0 \end{smallmatrix}\right)$

the choice is $\{1\}$ if $t \in E$ and $\{1, 2\}$ otherwise. Starting from $E = \mathbb{R}$ (diagonal ranking) and removing one untested ratio defeats any finite suite. □

PROPOSITION 12.5. *There are $2^{\mathfrak{c}}$ rules satisfying* N, F, S, *and* O *that agree on every binary matrix.*

Proof. Let $r_i = (M_{ii} - \min_j M_{ij})/(\max_j M_{ij} - \min_j M_{ij})$, with $r_i = 0$ for a constant row, and for $E \subseteq (0, 1)$ rank by $s_i = \mathbf{1}\{r_i \in E\}$. These rules satisfy N, F, S, and O, even under separate positive affine transformations of rows. On binary matrices every r_i is 0 or 1, so every score is zero; on $\left(\begin{smallmatrix} t & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ the scores are $(\mathbf{1}\{t \in E\}, 0, 0)$. □

Together with Corollary 11.2, these results give the following picture.

assumptions beyond A0	size of the family; certification
N, F, O	$2^{\mathfrak{c}}$; finite suites certify neither P nor D
N, F, B	$2^{\mathfrak{c}}$; finite suites do not certify D
N, F, S, O	$2^{\mathfrak{c}}$; all binary behaviour can leave the rule undetermined
N, F, C, O, Q, Mon	$2^{\mathfrak{c}}$; finite suites do not certify D (Section 11)
N, F, B, O	exactly three rules; one unequal-diagonal test distinguishes them

"The relevant types are few" is thus true in exactly one of these regimes, and there only conditionally: when principal binary independence and ordering are known to hold globally, a single binary test decides among the three possible rules. Finite observation never verifies those structural axioms themselves.

13. Mixed policies and evidential semantics

13.1. A canonical mixed extension

A pure causal matrix does not determine how prediction responds to a mixed policy, and continuity alone does not specify the extension. One precise extension takes the predictor's response to be linear in the policy and the action to be drawn independently of the response given the announced distribution:

$$\mu_p = \sum_i p_i \mu_i, \qquad g(p) = p^\top M, \qquad V(p) = p^\top M p,$$

where $g(p)_j$ is the causal value of option j under the beliefs induced by p. This differs from a predictor of the realised sample, which can instead yield the linear value $\sum_i p_i M_{ii}$. Define

D_m p maximises $p^\top M p$;

J_m every action in the support of p maximises $p^\top M$;

P_m p assigns probability zero to every weakly dominated column.

PROPOSITION 13.1. *In the canonical extension,* P_m *and* J_m *are jointly satisfiable on every matrix.*

Proof. For $0 < \eta < 1/|I|$ let $\Delta_\eta = \{p : p_i \geq \eta \text{ for all } i\}$. The correspondence $p \mapsto \arg\max_{q \in \Delta_\eta} p^\top M q$ on the compact convex set Δ_η has nonempty convex values and a closed graph, so Kakutani's theorem gives $p^\eta \in \arg\max_{q \in \Delta_\eta} (p^\eta)^\top M q$. If column a weakly dominates column b, full support makes $(p^\eta)^\top M e_a > (p^\eta)^\top M e_b$, so $p^\eta_b = \eta$; otherwise moving mass from b to a would improve the response. Let p be a limit point as $\eta \downarrow 0$. It assigns zero mass to every dominated column. For any q, put $q^\eta = (1 - |I|\eta)q + \eta \mathbf{1} \in \Delta_\eta$; then $(p^\eta)^\top M p^\eta \geq (p^\eta)^\top M q^\eta$, and in the limit $p^\top M p \geq p^\top M q$. Hence p is supported on best responses to itself. □

PROPOSITION 13.2. $\{\mathsf{P}_m, \mathsf{D}_m\}$ *and* $\{\mathsf{J}_m, \mathsf{D}_m\}$ *are inconsistent.*

Proof. For $M = \left(\begin{smallmatrix} 2 & 3 \\ 0 & 1 \end{smallmatrix}\right)$ and $p = (x, 1 - x)$, $V(p) = 1 + x$, so D_m selects $x = 1$, while column 2 dominates and is the unique best response to every p, so P_m and J_m select $x = 0$. □

PROPOSITION 13.3. *For rules that select a single lottery,* N *and* D_m *are inconsistent.*

Proof. For $M = I_2$, the swap fixes M, so a neutral single lottery is $(\frac{1}{2}, \frac{1}{2})$; but $V(p) = p_1^2 + p_2^2$ is maximised only at the two pure endpoints. □

A neutral *correspondence* of maximising lotteries exists; a single neutral maximising lottery need not. The pure consistency maps therefore cannot simply be relabelled as mixed maps.

13.2. Contraction for lotteries

For a rule L that selects one lottery, the natural strengthening of contraction is conditional-renormalisation consistency:

$$\mathsf{C}_m : \qquad L(M|_K) = \frac{L(M)|_K}{L(M)(K)} \quad \text{whenever } L(M)(K) > 0.$$

PROPOSITION 13.4. $\{\mathsf{N}, \mathsf{R}, \mathsf{C}_m, \mathsf{Q}\}$ *is a minimal inconsistent family for single-lottery rules.*

Proof. In the circulant extension of Theorem 9.3, neutrality forces the uniform lottery; conditioning forces the uniform lottery on every embedded normalised matrix; row translations then force the uniform lottery everywhere. For minimality, take the uniform distribution over the choice set of, respectively, the responsive fixed-label rule, diagonal ranking, row-range ranking, and universal indifference. The first two maximise a binary-independent weak order, so that $G(M|_K) = G(M) \cap K$ whenever the right-hand side is nonempty, which is exactly what C_m requires of uniform lotteries; row-range ranking is the witness that omits C_m. □

Support contraction alone is much weaker: $p_i \propto 1 + r_i/r_{\max}$, with r_i the row ranges and the uniform lottery when $r_{\max} = 0$, has full support everywhere, satisfies N, F, R, and is nonconstant. A complete mixed consistency map requires choosing among genuinely different output and contraction semantics; the results above are the extensions established under the stated ones.

13.3. Two evidential semantics

The matrix does not supply an evidential criterion until the conditioning experiment is identified. Two natural choices give different answers.

Reference-policy law. With a full-support reference ρ and $P(I = i, s) = \rho_i \mu_i(s)$, conditioning on the policy index i yields μ_i, and the evidential value of i is $\mathbb{E}[U(s, i) \mid I = i] = M_{ii}$. This semantics gives diagonal optimality, and it is the matrix form of $C6^K$ on single-choice problems.

Conditioning on the realised action under a known mixed policy. In the canonical extension the environment has law μ_p, and observing a positive-probability realised action does not change it, so its conditional value is $g_i(p)$. With an explicit completion for zero-probability alternatives, the resulting support-optimality condition is ratifiability.

The two condition on different variables under different joint laws, and neither can be silently substituted for the other. This is one precise sense in which "evidential decision theory" names a family rather than a criterion, as the literature on the tickle defence and on ratifiability has long made clear (Eells 1982; Jeffrey 1983).

14. Conclusion

The question was whether a decision theory can be optimal on every fair decision problem, and the expectation was that conflicting criteria of optimality would rule it out. The answer has three parts.

First, *ex-ante optimality is always attainable.* On every class of well-defined finite problems, choosing an ex-ante maximiser in each problem is a decision theory (Theorem 3.6). Because lookup tables count as decision theories, an impossibility for criteria that judge each problem separately is never more than a single problem on which their acceptable sets are disjoint (Lemma 3.7).

Second, *within a problem the conflicts are shallow.* Among ex-ante optimality, structural causal dominance, ratifiability, existence, commitment stability, and a reference-policy evidential criterion, every inconsistency has a core of one or two criteria, and the maximal consistent combinations have explicit total witnesses (Theorems 5.7 and 5.10). The central conflict is Nozick's. It is possible exactly when the law of the states preceding a choice depends on the prescription (Theorem 4.2); it survives imperfect prediction, randomisation, and every restriction on prediction targets short of excluding anticipation; and it obeys an exact trade-off between ex-ante and causal regret (Proposition 6.1). The conflict between dominance and commitment stability, by contrast, is confined on its witness to infallible prediction, whereas its

expectational counterpart persists whenever one-boxing has the higher ex-ante value (Proposition 5.11).

Third, *across problems, the depth depends on the semantics of menus.* On the causal matrix, with common affine invariance, the cross-problem axioms reduce to singleton and pairwise cores (Theorem 8.1). Row-translation invariance—evaluation insensitive to the baseline of each belief state—produces a minimal four-axiom impossibility: neutrality, row translations, and contraction force universal choice (Theorem 9.3), exactly on domains satisfying (DR) (Theorem 9.7), a condition that survives imperfect prediction, bounded action sets, and, with affine invariance, bounded payoffs. Principal binary independence, with neutrality, affine invariance, ordering, non-triviality, and monotonicity, characterises ex-ante evaluation (Corollary 11.2). Both results presuppose principal restriction, on which removing an option also removes the belief state it would induce. Under column or fixed-reference restriction, fixed-prior causal evaluation satisfies every axiom considered except diagonal optimality, self-sovereignty, and principal binary independence (Proposition 10.7), and contraction under any semantics is compatible with $2^{\mathfrak{c}}$ rules that do not maximise ex-ante value (Proposition 11.5).

The thesis that the paper can defend is therefore the following.

> *On any class of well-defined, fair problems containing a sufficiently predictive Newcomb problem, no assignment of policies satisfies both ex-ante optimality and causal dominance, although an ex-ante optimal assignment exists. Within problems, every inconsistency among the natural criteria has a core of size at most two. Across problems, on the unrestricted causal-matrix domain, principal binary independence together with neutrality, ordering, affine invariance, non-triviality, and monotonicity characterises ex-ante ranking, and principal contraction together with neutrality is incompatible with non-trivial, baseline-invariant choice on every domain satisfying (DR). Neither cross-problem result survives when a smaller menu retains the belief model of the larger one.*

Several questions remain open. The first is normative rather than mathematical: which semantics of menu restriction is the right one. Principal restriction is forced by the causal matrix once law-preserving removal is assumed, but an announced restriction may change a predictor's behaviour, and a fixed belief model is the natural reading of fixed-prior causal theories. The cross-problem impossibilities stand or fall with that choice, and the paper has shown only how they depend on it. The second is whether structural dominance and commitment stability are jointly satisfiable on a whole class of problems with imperfect prediction: Proposition 5.11 removes the Newcomb witness, and Proposition 5.12 decides dominance among commitments, but no general consistency theorem has been proved. The third concerns the reduced form. The causal matrix discards structural states and the local structure of multi-stage problems, so that column Pareto is not structural dominance and global ratifiability is not local ratifiability; a cross-problem theory on a richer reduced form might find obstructions that the matrix cannot express. The fourth concerns regularity:

TABLE 1. The principal results, with their status. "Not found" means not found in the literature searched; no priority is claimed.

result	where	status
ex-ante optimality vs dominance (Newcomb)	Prop. 4.1	Nozick (1969)
no ratifiable pure act (Death in Damascus)	Prop. 4.3	Gibbard and Harper (1978); ratifiability literature
universal ex-ante selection; pointwise reduction	Thm. 3.6, Lem. 3.7	elementary
characterisation of the dominance conflict	Thm. 4.2	not found
Bomb: ill-posedness and repair	Props. 4.4, 4.5	case from MacAskill (2019)
within-problem maps	Thms. 5.7, 5.10	not found
noisy commitment and couplings	Props. 5.11, 5.12	uses Strassen (1965)
regret trade-off	Prop. 6.1	not found
mixed existence under continuity	Thm. 6.3	Halpern (2020); Bell et al. (2021)
realisability of every causal matrix	Thm. 7.2	elementary
R as factorisation; R is not causal rationality	Thm. 7.8, Prop. 7.9	not found
Pareto vs self-sovereignty	Prop. 8.5	Briggs (2010)
pairwise reduction under F	Thm. 8.1	not found
circulant theorem; map with R	Thms. 9.3, 9.6	not found; ingredients in Fink and Ruiz (1986), Brandt et al. (2016)
exact domain criterion (DR); physical restrictions	Thm. 9.7, Props. 9.8–9.11	not found
menu semantics; weight preservation; column maps	Sec. 10	not found; cf. Gallow (2026)
binary characterisation of ex-ante ranking	Thm. 11.1, Cor. 11.2	core in Podgorski (2022)
contraction without the diagonal; $2^{\mathfrak{c}}$ families	Thm. 11.4, Props. 11.5–12.5	not found
mixed and evidential extensions	Sec. 13	not found

the cardinality results use the freedom to define rules arbitrarily on orbits, and the classification of rules subject to continuity or measurability is open. The fifth is a complete consistency map for mixed policies, which requires a choice among several inequivalent semantics for output, prediction, and contraction. Finally, the status in the literature of the circulant theorem and of the consistency maps has been checked only by a bounded search; their ingredients are classical, and an earlier statement may exist.

A. Register of stipulations

The results are theorems about the definitions displayed in the text. The following choices are made there, and each could be made differently; where a different choice is known to change a result, the section that discusses it is given.

1. Policies are deterministic unless a mixed variant is stated (Sections 6, 13).
2. Every admissible policy induces a well-defined probability law (Definition 2.3); self-referential problems that lack one are excluded (Remark 7.3).
3. No information set occurs twice on a path (Definition 2.1).
4. Structural environment plans, including responses on branches not taken, are part of the problem (Definition 2.2); how outcomes on alternative branches are coupled is fixed by them.
5. Causal deviations freeze the baseline distribution over plans, and a local deviation keeps the policy's other prescriptions (definition of Q_π).
6. Dominance quantifies over structural states, including those of probability zero under the evaluated policy (Section 3); with supported states only, C2 becomes unsatisfiable in Death in Damascus.
7. Positive reach is measured under the evaluated policy.
8. Predictor error is given by a kernel; the Newcomb calculations use symmetric error.
9. A predictor targets specified policy coordinates; "the same computation" is represented by a stipulated dependence (Sections 4.2, 4.4).
10. Commitment is irrevocable, precedes the environment's response, offers the full policy set, and uses a specified coupling and structural state set (Definition 5.1, Section 5.4).
11. Commitment extensions are tagged and well founded, and the class is closed under them.
12. C5 is the totality already built into the definition of a theory.
13. The evidential criterion uses the reference-policy law and the null-event convention displayed in Section 5.3; other evidential semantics change the map (Section 13.3).

14. Mixed variants specify sampling, predictor access, information, counterfactual coupling, and continuity (Sections 6.3, 13).
15. Cross-problem rules output a reflexive relation whose maximal elements are required to be nonempty (Definition 7.4); neutrality acts by bijections between label sets, and invariances act on the relation.
16. Ordering concerns the output relation at each input, not rationalisability of each choice set.
17. Menu restriction is principal unless stated otherwise; principal restriction is forced by law-preserving removal, and column and fixed-reference restriction require additional data (Section 10).
18. Non-triviality is existential unless a universe is specified (Theorem 10.6).
19. On restricted domains, an invariance applies when both a matrix and its transform are admissible (Section 9).
20. Equivalence of problems and parameterisation of rules are made precise before taxonomy claims are stated (Section 12).

⁂

References

Ahmed, A. (2012). "Push the Button". *Philosophy of Science* 79.3, pp. 386–395.

Ahmed, A. (2014a). "Dicing with Death". *Analysis* 74.4, pp. 587–592.

Ahmed, A. (2014b). *Evidence, Decision and Causality*. Cambridge: Cambridge University Press.

Ahmed, A., ed. (2018). *Newcomb's Problem*. Classic Philosophical Arguments. Cambridge: Cambridge University Press.

Arntzenius, F. (2008). "No Regrets, or: Edith Piaf Revamps Decision Theory". *Erkenntnis* 68.2, pp. 277–297.

Arntzenius, F., A. Elga, and J. Hawthorne (2004). "Bayesianism, Infinite Decisions, and Binding". *Mind* 113.450, pp. 251–283.

Arrow, K. J. (1951). *Social Choice and Individual Values*. Cowles Commission Monograph 12. New York: John Wiley & Sons.

Arrow, K. J. (1959). "Rational Choice Functions and Orderings". *Economica*. New ser. 26.102, pp. 121–127.

Aumann, R. J., S. Hart, and M. Perry (1997). "The Absent-Minded Driver". *Games and Economic Behavior* 20.1, pp. 102–116.

Bales, A. (2018). "Richness and Rationality: Causal Decision Theory and the WAR Argument". *Synthese* 195.1, pp. 259–267.

Basu Mallik, A. and M. Bhattacharya (Dec. 2026). "Consistency and Social Choice: Possibilities and Impossibilities". *Mathematical Social Sciences* 144, 102582.

Bell, J., L. Linsefors, C. Oesterheld, and J. Skalse (2021). "Reinforcement Learning in Newcomb-like Environments". In: *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*. Ed. by M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, and J. Wortman Vaughan. Vol. 34. Curran Associates, Inc., pp. 22146–22157.

Blair, D. H., G. Bordes, J. S. Kelly, and K. Suzumura (1976). "Impossibility Theorems without Collective Rationality". *Journal of Economic Theory* 13.3, pp. 361–379.

Bolker, E. D. (1966). "Functions Resembling Quotients of Measures". *Transactions of the American Mathematical Society* 124.2, pp. 292–312.

Bolker, E. D. (1967). "A Simultaneous Axiomatization of Utility and Subjective Probability". *Philosophy of Science* 34.4, pp. 333–340.

Brandt, F. (2011). "Minimal Stable Sets in Tournaments". *Journal of Economic Theory* 146.4, pp. 1481–1499. arXiv: 0803.2138.

Brandt, F., M. Brill, and P. Harrenstein (2016). "Tournament Solutions". In: *Handbook of Computational Social Choice*. Ed. by F. Brandt, V. Conitzer, U. Endriss, J. Lang, and A. D. Procaccia. Cambridge: Cambridge University Press. Chap. 3, pp. 57–84.

Brandt, F. and P. Harrenstein (2011). "Set-Rationalizable Choice and Self-Stability". *Journal of Economic Theory* 146.4, pp. 1721–1731. arXiv: 0910.3580.

Briggs, R. (2010). "Decision-Theoretic Paradoxes as Voting Paradoxes". *The Philosophical Review* 119.1, pp. 1–30.

Chernoff, H. (1954). "Rational Selection of Decision Functions". *Econometrica* 22.4, pp. 422–443.

Cooper, E., C. Oesterheld, and V. Conitzer (Nov. 7, 2024). *Can CDT Rationalise the Ex Ante Optimal Policy via Modified Anthropics?* Version 2. arXiv: 2411.04462 [cs.AI].

Dai, W. (Aug. 13, 2009). *Towards a New Decision Theory*. LessWrong. URL: https://www.lesswrong.com/posts/de3xjFaACCAk6imzv/towards-a-new-decision-theory (visited on 10/19/2026).

Eells, E. (1982). *Rational Decision and Causality*. Cambridge: Cambridge University Press.

Egan, A. (2007). "Some Counterexamples to Causal Decision Theory". *The Philosophical Review* 116.1, pp. 93–114.

Everitt, T., J. Leike, and M. Hutter (2015). "Sequential Extensions of Causal and Evidential Decision Theory". In: *Algorithmic Decision Theory: 4th International Conference, ADT 2015, Lexington, KY, USA, September 27–30, 2015, Proceedings*. Ed. by T. Walsh. Vol. 9346. Lecture Notes in Computer Science. Cham: Springer, pp. 205–221.

Fink, J. F. and S. Ruiz (1986). "Every Graph Is an Induced Isopart of a Circulant". *Czechoslovak Mathematical Journal* 36.2, pp. 172–176.

Gallow, J. D. (2020). "The Causal Decision Theorist's Guide to Managing the News". *The Journal of Philosophy* 117.3, pp. 117–149.

Gallow, J. D. (2026). "Comparativism About Instrumental Value". *Philosophy and Phenomenological Research* 113.2, pp. 346–363.

Gauthier, D. (1986). *Morals by Agreement*. Oxford: Clarendon Press.

Gauthier, D. (1994). "Assure and Threaten". *Ethics* 104.4, pp. 690–721.

Gibbard, A. and W. L. Harper (1978). "Counterfactuals and Two Kinds of Expected Utility". In: *Foundations and Applications of Decision Theory*. Ed. by C. A. Hooker, J. J. Leach, and E. F. McClennen. Vol. 1. Dordrecht: D. Reidel, pp. 125–162.

Halpern, J. Y. (2020). "A Note on the Existence of Ratifiable Acts". *The Review of Symbolic Logic* 13.3, pp. 503–508.

Hammond, P. J. (1988). "Consequentialist Foundations for Expected Utility". *Theory and Decision* 25.1, pp. 25–78.

Hare, C. and B. Hedden (2016). "Self-Reinforcing and Self-Frustrating Decisions". *Noûs* 50.3, pp. 604–628.

Harper, W. (1986). "Mixed Strategies and Ratifiability in Causal Decision Theory". *Erkenntnis* 24.1, pp. 25–36.

Jeffrey, R. C. (1965). *The Logic of Decision*. New York: McGraw-Hill.

Jeffrey, R. C. (1981). "The Logic of Decision Defended". *Synthese* 48.3, pp. 473–492.

Jeffrey, R. C. (1983). *The Logic of Decision*. 2nd ed. Chicago: University of Chicago Press.
Joyce, J. M. (1999). *The Foundations of Causal Decision Theory*. Cambridge: Cambridge University Press.
Joyce, J. M. (2012). "Regret and Instability in Causal Decision Theory". *Synthese* 187.1, pp. 123–145.
Joyce, J. M. (2018). "Deliberation and Stability in Newcomb Problems and Pseudo-Newcomb Problems". In: *Newcomb's Problem*. Ed. by A. Ahmed. Classic Philosophical Arguments. Cambridge: Cambridge University Press, pp. 138–159.
Kakutani, S. (1941). "A Generalization of Brouwer's Fixed Point Theorem". *Duke Mathematical Journal* 8.3, pp. 457–459.
Kuhn, H. W. (1953). "Extensive Games and the Problem of Information". In: *Contributions to the Theory of Games, Volume II*. Ed. by H. W. Kuhn and A. W. Tucker. Annals of Mathematics Studies 28. Princeton, NJ: Princeton University Press, pp. 193–216.
Laslier, J.-F. (1997). *Tournament Solutions and Majority Voting*. Studies in Economic Theory. Berlin, Heidelberg: Springer.
Levinstein, B. A. and N. Soares (2020). "Cheating Death in Damascus". *The Journal of Philosophy* 117.5, pp. 237–266.
Lewis, D. (1979). "Prisoners' Dilemma Is a Newcomb Problem". *Philosophy & Public Affairs* 8.3, pp. 235–240.
Lewis, D. (1981a). "'Why Ain'cha Rich?'" *Noûs* 15.3, pp. 377–380.
Lewis, D. (1981b). "Causal Decision Theory". *Australasian Journal of Philosophy* 59.1, pp. 5–30.
MacAskill, W. (Sept. 13, 2019). *A Critique of Functional Decision Theory*. LessWrong (cross-posted to the AI Alignment Forum). URL: https://www.lesswrong.com/posts/ySLYSsNeFL5CoAQzN/a-critique-of-functional-decision-theory (visited on 10/19/2026).
MacDermott, M., T. Everitt, and F. Belardinelli (July 20, 2023). *Characterising Decision Theories with Mechanised Causal Graphs*. arXiv: 2307.10987 [cs.AI].
McClennen, E. F. (1990). *Rationality and Dynamic Choice: Foundational Explorations*. Cambridge: Cambridge University Press.
Moulin, H. (1986). "Choosing from a Tournament". *Social Choice and Welfare* 3.4, pp. 271–291.
Nesov, V. (2009). *Counterfactual Mugging*. LessWrong. URL: https://www.lesswrong.com/posts/mg6jDEuQEjBGtibX7/counterfactual-mugging.
Nozick, R. (1969). "Newcomb's Problem and Two Principles of Choice". In: *Essays in Honor of Carl G. Hempel*. Ed. by N. Rescher. Dordrecht: D. Reidel, pp. 114–146.
Oesterheld, C. and V. Conitzer (2021). "Extracting Money from Causal Decision Theorists". *The Philosophical Quarterly* 71.4, pp. 701–716.
Oesterheld, C. and V. Conitzer (Jan. 5, 2024). "Can de se Choice Be Ex Ante Reasonable in Games of Imperfect Recall? A Complete Analysis".
Piccione, M. and A. Rubinstein (1997). "On the Interpretation of Decision Problems with Imperfect Recall". *Games and Economic Behavior* 20.1, pp. 3–24.
Podgorski, A. (2022). "Tournament Decision Theory". *Noûs* 56.1, pp. 176–203.
Rabinowicz, W. (1988). "Ratifiability and Stability". In: *Decision, Probability, and Utility: Selected Readings*. Ed. by P. Gärdenfors and N.-E. Sahlin. Cambridge: Cambridge University Press, pp. 406–425.
Ramsey, F. P. (1931). "Truth and Probability". In: *The Foundations of Mathematics and Other Logical Essays*. Ed. by R. B. Braithwaite. London: Kegan Paul, Trench, Trubner & Co., pp. 156–198.
Richter, R. (1984). "Rationality Revisited". *Australasian Journal of Philosophy* 62.4, pp. 392–403.
Savage, L. J. (1954). *The Foundations of Statistics*. New York: Wiley.

Sen, A. K. (1970). *Collective Choice and Social Welfare*. Mathematical Economics Texts 5. San Francisco: Holden-Day.
Sen, A. K. (1971). "Choice Functions and Revealed Preference". *The Review of Economic Studies* 38.3, pp. 307–317.
Sen, A. K. (1977). "Social Choice Theory: A Re-Examination". *Econometrica* 45.1, pp. 53–89.
Sen, A. K. (1993). "Internal Consistency of Choice". *Econometrica* 61.3, pp. 495–521.
Skalse, J. (Jan. 1, 2021). *A General Counterexample to Any Decision Theory and Some Responses*. arXiv: 2101.00280 [cs.AI].
Skyrms, B. (1980). *Causal Necessity: A Pragmatic Investigation of the Necessity of Laws*. New Haven, CT: Yale University Press.
Skyrms, B. (1982). "Causal Decision Theory". *The Journal of Philosophy* 79.11, pp. 695–711.
Skyrms, B. (1990a). "Ratifiability and the Logic of Decision". *Midwest Studies in Philosophy* 15, pp. 44–56.
Skyrms, B. (1990b). *The Dynamics of Rational Deliberation*. Cambridge, MA: Harvard University Press.
Soares, N. and B. Fallenstein (July 7, 2015). *Toward Idealized Decision Theory*. arXiv: 1507.01986 [cs.AI].
Sobel, J. H. (1994). *Taking Chances: Essays on Rational Choice*. Cambridge: Cambridge University Press.
Spencer, J. and I. Wells (2019). "Why Take Both Boxes?" *Philosophy and Phenomenological Research* 99.1, pp. 27–48.
Spohn, W. (1977). "Where Luce and Krantz Do Really Generalize Savage's Decision Model". *Erkenntnis* 11.1, pp. 113–134.
Spohn, W. (2012). "Reversing 30 Years of Discussion: Why Causal Decision Theorists Should One-Box". *Synthese* 187.1, pp. 95–122.
Strassen, V. (1965). "The Existence of Probability Measures with Given Marginals". *The Annals of Mathematical Statistics* 36.2, pp. 423–439.
Von Neumann, J. and O. Morgenstern (1944). *Theory of Games and Economic Behavior*. Princeton, NJ: Princeton University Press.
Wedgwood, R. (2013). "Gandalf's Solution to the Newcomb Problem". *Synthese* 190.14, pp. 2643–2675.
Weirich, P. (1985). "Decision Instability". *Australasian Journal of Philosophy* 63.4, pp. 465–472.
Weirich, P. (2024). "Causal Decision Theory". In: *The Stanford Encyclopedia of Philosophy*. Ed. by E. N. Zalta and U. Nodelman.
Wells, I. (2019). "Equal Opportunity and Newcomb's Problem". *Mind* 128.510, pp. 429–457.
Wolpert, D. H. and G. Benford (2013). "The Lesson of Newcomb's Paradox". *Synthese* 190.9, pp. 1637–1646.
Yudkowsky, E. (2010). *Timeless Decision Theory*. Tech. rep. San Francisco, CA: The Singularity Institute (now Machine Intelligence Research Institute).
Yudkowsky, E. and N. Soares (Oct. 13, 2017). *Functional Decision Theory: A New Theory of Instrumental Rationality*. Version 2. arXiv: 1710.05060 [cs.AI].